

A PROJECT REPORT
on
**AN ENSEMBLE LEARNING APPROACH FOR
IMPROVED SLEEP DISORDER PREDICTION**

Submitted in partial fulfillment for the award of the degree of

Master of Technology
in
Artificial Intelligence and Data Science

Submitted by

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Department of Computer Science and Engineering

DECLARATION BY THE CANDIDATE

I hereby declare that the Report entitled “**AN ENSEMBLE LEARNING APPROACH FOR IMPROVED SLEEP DISORDER PREDICTION**” submitted by **P. Bhargavi (24MT8202)** to the Department of Computer Science and Engineering, School of Engineering and Technology, Sri Padmavati Mahila Visvavidyalayam, Tirupati in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Artificial Intelligence and Data Science** is a record of bonafide work carried out by me under the supervision of **Ms. G. Sreehitha, Assistant Professor** and **Ms. G. Radhika, Assistant Professor**. I further declare that the work reported has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma of this university or any other institute or university.

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BONAFIDE CERTIFICATE

This is to certify that the report entitled “**AN ENSEMBLE LEARNING APPROACH FOR IMPROVED SLEEP DISORDER PREDICTION**” submitted by **P. Bhargavi (24MT8202)** to the Department of Computer Science and Engineering, School of Engineering and Technology, Sri Padmavati Mahila Visvavidyalayam, Tirupati in partial fulfillment of the requirement for the award of the degree of **M. Tech in Artificial Intelligence and Data Science** is a record of bonafide work carried out by her under our guidance. The work fulfills the requirements as per the regulations of this University and in my opinion, meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

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ABSTRACT

Sleep disorders significantly impact both physical and mental health, leading to issues such as fatigue, reduced productivity, anxiety, and other long-term health complications. Traditional diagnostic methods like Polysomnography (PSG), although accurate, are often expensive, time-consuming, uncomfortable for patients, and not easily accessible to the general population. This creates a need for a more convenient, cost-effective, and widely accessible solution for early detection of sleep disorders. To address this problem, this study proposes a machine learning-based Sleep Disorder Prediction System that allows users to assess their sleep health through a simple and user-friendly interface. The system collects user inputs such as age, gender, occupation, sleep duration, caffeine intake, blood pressure, heart rate, BMI category, snoring frequency, working hours, and other health-related data to predict possible sleep disorders. The backend of the system is developed using Python, and multiple machine learning algorithms are applied to improve prediction accuracy. Instead of relying on individual models such as K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest, the system utilizes ensemble learning techniques, including Voting Classifiers or Stacking methods. These techniques combine the strengths of multiple models to produce more reliable and accurate predictions of sleep disorders. The model is integrated into a mobile application to provide a fast, cost-effective, and user-friendly tool for detecting sleep disorders. This approach offers a practical alternative to traditional tests and may help improve patient outcomes.

Keywords: Sleep Disorder, KNN, SVM, Random Forest, Voting Classifier, Mobile Application.

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CHAPTER 1

INTRODUCTION

1.1 General Introduction

Sleep is essential for maintaining good physical health, mental stability, and daily performance, but traditional diagnostic methods like Polysomnography (PSG) are costly, uncomfortable, and not easily available. This makes diagnosis difficult, especially for people in rural or low-resource areas. In many cases, individuals are unaware of their sleep-related issues until they become severe, leading to long-term health complications such as stress, cardiovascular problems, and reduced work efficiency. The lack of awareness and limited access to diagnostic facilities further increases the risk of untreated sleep disorders. To address these challenges, this project uses machine learning to provide a simpler and more accessible method for identifying sleep disorders. By analyzing basic health and lifestyle information, machine learning models can quickly predict whether a person is likely to have a disorder.

The system collects various input parameters such as age, occupation, sleep duration, stress levels, caffeine intake, blood pressure, heart rate, BMI category, and other lifestyle factors. These inputs are used to train and evaluate the model so that it can recognize patterns and correlations associated with different types of sleep disorders. While individual models such as KNN, SVM and Random Forest provide good results, they often vary in accuracy and stability. Therefore, our project uses ensemble learning techniques specifically Stacking or Voting Classifiers which combine multiple models to produce more reliable and accurate predictions. Ensemble methods help in reducing overfitting and improve the overall performance of the system by leveraging the strengths of different algorithms. This results in better generalization and more consistent predictions across different types of input data. The system is enabling it to learn important patterns related to sleep health. To increase usability, the final model is integrated into a mobile application, allowing users to enter their data and receive instant predictions anytime, anywhere be affected by a variety of environmental factors, including temperature, sunshine, nutrition, and humidity. An efficient automated system is needed to identify and separate flower species. In addition, the system is also deployed as a web application using an online hosting platform, allowing users to access it through a simple link

without installing any software. The mobile application (APK) is developed by converting the web application into a portable format, ensuring accessibility across different devices. This integration of web and mobile platforms enhances user convenience and ensures that the system can be used by a wider audience. Overall, the proposed solution provides a practical, efficient, and user-friendly approach for early detection and monitoring of sleep disorders.

1.2 Thesis Statement

Sleep disorders are increasingly affecting people's health and daily life, but early diagnosis remains a major challenge. Traditional diagnostic methods such as Polysomnography (PSG) are expensive, time-consuming, and require specialized medical facilities, making them inaccessible to many individuals, especially in rural and low-resource areas. There is no simple and easily available system that allows users to assess their sleep health using basic personal and lifestyle information. As a result, many people remain undiagnosed or delay seeking medical help until the condition becomes severe. Therefore, there is a need to develop a cost-effective, accessible, and user-friendly system that can predict sleep disorders at an early stage using machine learning techniques. The system should be easily accessible through a web link and mobile application, enabling users to get quick predictions anytime and anywhere.

1.3 Objectives of the Proposed Work

- To develop an efficient and user-friendly system for predicting sleep disorders using machine learning techniques.
- To provide an accessible and cost-effective solution for early detection of sleep-related issues.
- To analyze user input data such as lifestyle habits, health parameters, and sleep patterns for accurate prediction.
- To assist individuals in identifying potential sleep disorders at an early stage and encourage timely medical consultation.
- To improve awareness about sleep health and its impact on overall well-being. To design a system that delivers quick and reliable results with minimal user effort.

1.4 Report Organization

The report is systematically organized into multiple chapters to provide a comprehensive understanding of the proposed system, “An Ensemble Learning Approach for Improved Sleep Disorder Prediction.” Each chapter focuses on a specific aspect of the project, ranging from conceptual background to implementation and evaluation.

Chapter 2: Literature Survey

This chapter reviews existing research works and methodologies used for sleep disorder detection. It analyzes various machine learning models such as KNN, SVM, and Random Forest, and identifies gaps in existing systems that justify the need for an improved ensemble-based solution.

Chapter 3: Proposed System Overview

This chapter presents a detailed analysis of the existing system and its limitations, such as dependency on expensive medical tests like polysomnography. It also introduces the proposed system, highlighting its advantages such as cost-effectiveness, accessibility, and improved prediction accuracy using real-time user input data.

Chapter 4: Proposed System Analysis

This chapter explains the overall architecture of the system. It includes UML diagrams such as use case diagrams, class diagrams, sequence diagrams, and flow diagrams. It also describes data flow, module design, and interactions between frontend, backend, and machine learning components.

Chapter 5: Results and Discussion

This chapter presents the experimental results obtained from the system. It includes performance metrics such as accuracy, precision, recall, and F1-score. A comparison between individual models and the ensemble model is also discussed to demonstrate performance improvement.

Chapter 6: Conclusion and Future Work

This chapter summarizes the overall work and highlights the key achievements of the project. It also suggests possible future enhancements, such as integrating wearable device data, improving model accuracy, and deploying the system as a mobile or cloud-based application.

CHAPTER 2

LITERATURE SURVEY

2.1 Survey of the Existing Models

Muhammad Mostafa Monowar, S. M. Nuruzzaman Nobel, Maharin Afroj, “Advanced Sleep Disorder Detection Using Multi-Layered Ensemble Learning and Advanced Data Balancing Techniques”, 2025 [1]. The existing system focuses on detecting sleep disorders using single machine learning models; however, their accuracy and stability often vary, making them less reliable for consistent predictions. Although the proposed multi-layered ensemble approach improves performance compared to individual models, it introduces several challenges. The model is computationally intensive, requiring high processing power and longer training time, which makes it less suitable for real-time or resource-constrained environments. Additionally, there is a higher risk of overfitting due to the complexity of the ensemble techniques, especially when trained on limited datasets. Another significant limitation is the lack of validation using diverse external datasets, which affects the model’s ability to generalize well in real-world scenarios. Furthermore, the system may face difficulties in scalability, integration with practical healthcare systems, and handling noisy or incomplete user data, thereby limiting its widespread deployment and practical usability.

MD Kamal Rasel, Taanzina Afroz Rimi, Mahmuda Rawnak Jahan, Shayla Sharmin, “Improving Sleep Disorder Diagnosis Through Optimized Machine Learning Approaches”[2]. The primary limitation of this study is the use of a very small and limited dataset, which significantly reduces the reliability and real-world accuracy of the developed models. Due to insufficient data, the model may not capture diverse sleep patterns and health conditions, leading to poor generalization when applied to a larger population. The study also lacks validation with external or real-time datasets, which further limits its practical applicability. Moreover, the performance of the optimized machine learning techniques may not remain consistent across different demographic groups. The system may also struggle with handling noisy or incomplete data, and its scalability for large-scale deployment remains uncertain, thereby restricting its effectiveness in real-world healthcare applications.

C. Tran, Y. Wijesuriya, R. Thuraisingham, A. Craig, & H. Nguyen, "Deep Learning for Classification of Sleep Stages," 2019 [3]. This study focuses on the application of deep learning techniques for the classification of sleep stages using EEG data. The authors highlight the ability of deep learning models to effectively handle complex and high-dimensional sleep data, resulting in high classification accuracy. Their approach demonstrates significant improvements over traditional machine learning methods, providing a strong foundation for applying deep learning in sleep disorder classification. Additionally, the study shows that automated feature extraction in deep learning reduces the need for manual intervention, improving efficiency and consistency. The model is capable of identifying subtle patterns in EEG signals, which enhances the precision of sleep stage detection. However, the approach requires large volumes of labeled data and high computational resources for training. The study also indicates challenges in real-time implementation and dependency on high-quality EEG signals, which may limit its practical usage outside controlled clinical environments.

E. Alickovic & A. Subasi, "Ensemble SVM Method for Automatic Sleep Stage Classification," IEEE Transactions on Instrumentation and Measurement, 2018 [4]. This paper discusses the use of ensemble Support Vector Machine (SVM) methods for automatic sleep stage classification. The authors address the challenges of accurately classifying sleep stages and demonstrate the effectiveness of ensemble techniques in improving overall classification accuracy compared to single SVM models. Their approach combines multiple SVM classifiers to enhance robustness and reduce misclassification errors. The study highlights that ensemble learning can better handle variability in physiological signals such as EEG, leading to more stable and reliable predictions. Additionally, the method improves generalization performance and reduces the impact of noise in the data. However, the approach increases computational complexity and requires careful selection of parameters and model combinations. The study also indicates that the performance depends on the quality of input data and may require preprocessing steps for optimal results, which can add to system complexity.

M. J. Sun, Z. F. Wu, & X. B. Lu, "Sleep Apnea Detection Based on Time and Frequency Domain Analysis of ECG and SpO2 Signals," 2020 [5]. The authors propose a method for detecting sleep apnea using time and frequency domain analysis of ECG and SpO2 signals. By applying machine

learning algorithms to these physiological signals, they achieve high accuracy in identifying sleep apnea events. This research highlights the importance of combining multiple physiological data sources to improve detection performance. The use of both time-domain and frequency-domain features allows the model to capture detailed patterns and variations in heart rate and oxygen saturation levels, leading to more precise predictions. Additionally, advanced signal processing techniques help in reducing noise and improving the quality of input data. The study also demonstrates that integrating ECG and SpO2 signals enhances the reliability of the system compared to using a single data source. However, the approach requires continuous monitoring devices and high-quality signal acquisition, which may not always be feasible in non-clinical environments. Furthermore, the system may involve higher computational complexity and dependency on proper preprocessing methods, which can affect real-time implementation and scalability.

T. Radha, V. S. Kumar, & S. Pradeep, "Classification of Sleep Disorders Using Machine Learning Algorithms," *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*, 2019 [6]. This study explores the use of various machine learning algorithms for the classification of sleep disorders based on polysomnography (PSG) data. The authors compare the performance of different models, including decision trees, random forests, and neural networks, highlighting the advantages and limitations of each approach. Their results demonstrate the strong potential of machine learning techniques in improving the accuracy and efficiency of sleep disorder diagnosis. Additionally, the study shows that ensemble methods like random forests provide better performance due to their ability to reduce overfitting and handle complex data patterns. Neural networks, on the other hand, are effective in capturing non-linear relationships in PSG data, leading to improved classification results. The research also emphasizes the importance of feature selection and data preprocessing in enhancing model performance. However, the study indicates that the accuracy of these models depends heavily on the quality and size of the dataset. Furthermore, the use of PSG data requires specialized equipment and clinical settings, which may limit large-scale or real-time applications. The computational complexity of advanced models and the need for expert interpretation can also pose challenges for practical deployment.

K. Vuppalapati, V. Guddeti, & P. Prasad, "Sleep Disorder Classification Using EEG Signal Analysis and Machine Learning," 2021 [7]. The authors present a comprehensive approach to sleep disorder classification using EEG signal analysis and machine learning techniques. By extracting relevant features from EEG signals and applying classifiers such as SVM and deep neural networks, they achieve high classification accuracy. This research emphasizes the significance of feature extraction and model selection in the context of sleep disorder classification.

Raed Alazaidah et al., "Potential of Machine Learning for Predicting Sleep Disorders: A Comprehensive Analysis of Regression and Classification Models," 2023 [8]. This study provides a comprehensive analysis of various machine learning models used for predicting sleep disorders. The authors evaluate both regression and classification techniques to understand their effectiveness in diagnosing sleep-related conditions. The research highlights that machine learning models can significantly improve prediction accuracy when trained on well-structured datasets. It also emphasizes the importance of feature selection and data preprocessing in enhancing model performance. However, the study indicates that model performance varies depending on dataset quality and algorithm selection.

S. Gawhale et al., "EEG Signal Processing for the Identification of Sleeping Disorder Using Hybrid Deep Learning with Ensemble Machine Learning Classifier," 2023 [9]. This study proposes a hybrid approach combining deep learning and ensemble machine learning techniques for detecting sleep disorders using EEG signals. The model improves classification accuracy by leveraging both feature extraction and ensemble learning strategies. The research highlights the effectiveness of hybrid models in handling complex EEG data. However, it also notes that such models require high computational resources and careful tuning for optimal performance.

Seenivasan et al., "Sleep Disorder Classification Using AI: A Machine Learning Approach," 2025 [10]. This paper presents an AI-based system for automatically classifying sleep disorders using EEG signal analysis. The model extracts features such as frequency, amplitude, and wave patterns to identify sleep-related abnormalities. The study demonstrates that automation reduces manual effort and improves efficiency in diagnosis. However, the system depends on accurate signal acquisition and may face challenges with noisy data.

2.2 Gaps identified in the Survey

The analysis of existing sleep disorder prediction systems reveals several critical limitations in terms of methodology, usability, and real-world applicability. These gaps motivate the need for a more advanced and efficient solution.

Dependence on Clinical and Hardware-Based Data

Most traditional systems rely heavily on clinical data such as Polysomnography (PSG) or wearable sensors. These methods require continuous monitoring using expensive devices, limiting scalability and accessibility for common users.

Lack of Real-Time Prediction Capability

Many existing models are trained and tested only on offline datasets. They do not support real-time prediction based on user-provided inputs, which reduces their practical usability in everyday scenarios.

Imbalanced and Limited Datasets

Several research works use small or imbalanced datasets, where certain sleep disorders are underrepresented. This leads to biased predictions and reduced generalization capability of the models.

Overfitting and Poor Generalization

Single models like Decision Trees or KNN may overfit training data and fail to perform well on unseen data. This affects the reliability of predictions in real-world applications.

Absence of Hybrid or Ensemble Models

Many systems fail to combine multiple machine learning models. Without ensemble techniques like Voting or Stacking, they miss the opportunity to improve accuracy, reduce variance, and enhance robustness.

Lack of Feature Engineering and Optimization

Existing studies often do not focus on proper feature selection, scaling, or transformation

techniques. Poor feature engineering can significantly reduce model performance.

Minimal Consideration of Lifestyle Factors

Important attributes such as stress levels, daily activity, BMI, and age are sometimes ignored. This results in incomplete analysis and less accurate predictions.

Lack of Explainability (Black Box Models)

Some advanced models do not provide explanations for their predictions. This reduces trust among users, especially in healthcare-related applications.

No Personalization of Predictions

Most systems provide generalized predictions and do not adapt to individual user behavior or health conditions, limiting their effectiveness.

Weak User Interface and Interaction Design

Existing systems are often not designed with user experience in mind. Lack of intuitive UI makes it difficult for users to input data and interpret results.

Limited Deployment and Scalability

Many research models remain theoretical or prototype-level and are not deployed as scalable web or cloud-based applications.

Inadequate Evaluation Metrics

Some studies evaluate models only based on accuracy, ignoring other important metrics like precision, recall, F1-score, and confusion matrix, which are crucial for healthcare predictions.

Lack of Continuous Learning Mechanism

Most systems are static and do not update or improve over time based on new user data, limiting long-term performance improvement.

Lack of Data Preprocessing Standardization

Many existing works do not clearly define preprocessing steps such as handling missing values, normalization, or noise removal. This inconsistency affects model performance and reproducibility.

Absence of Cross-Validation Techniques

Some studies rely on simple train-test splits instead of robust validation methods like k-fold cross-validation, leading to unreliable performance evaluation.

Poor Handling of Noisy or Incomplete Data

Real-world user input can be incomplete or inconsistent, but most systems are not designed to handle such imperfect data effectively.

Limited Comparison with Baseline Models

Existing research often fails to compare proposed models with strong baseline algorithms, making it difficult to validate improvements.

High Computational Complexity

Some advanced models require high processing power and time, making them unsuitable for real-time or low-resource environments.

Lack of Lightweight Models for Deployment

Many systems are not optimized for deployment on web or mobile platforms due to heavy model size or complexity.

Security and Privacy Concerns Not Addressed

User health data is sensitive, yet many systems do not implement proper data security, encryption, or privacy-preserving techniques.

CHAPTER 3

OVERVIEW OF THE PROPOSED SYSTEM

3.1 Introduction

The proposed system aims to develop an efficient and user-friendly solution for predicting sleep disorders using advanced machine learning techniques. Sleep disorders such as insomnia, sleep apnea, and restless sleep significantly impact an individual's physical and mental health. Traditional diagnostic methods like Polysomnography (PSG) are often expensive, time-consuming, and require clinical supervision, making them less accessible for regular monitoring. To overcome these limitations, the proposed system utilizes an ensemble learning approach, which combines multiple machine learning algorithms such as K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest. By integrating these models using techniques like Voting or Stacking classifiers, the system improves prediction accuracy, stability, and reliability compared to individual models.

The system is designed to accept real-time user input data, including parameters such as sleep duration, heart rate, stress level, BMI, and age. This input is processed through a backend machine learning model, which analyzes patterns and predicts the likelihood of a sleep disorder. The results are then displayed to the user through a simple and interactive interface. Additionally, the system is implemented as a web-based application, ensuring easy accessibility and usability for a wide range of users. It eliminates the need for specialized medical equipment and enables early detection of sleep-related issues, allowing users to take preventive measures. Overall, the proposed system provides a cost-effective, accurate, and scalable solution for sleep disorder prediction by leveraging ensemble learning techniques and real-time data processing.

3.2 Architecture

The architecture of the proposed Sleep Disorder Prediction System is designed in a structured and modular way to ensure efficient data processing and accurate prediction. The system begins with the user interface (Sleep Disorder Prediction App), where users enter real-time health data such as sleep duration, heart rate, stress level, BMI, and age. This input is sent to the system through an

API Gateway, which acts as a communication bridge between the frontend and backend. The request is then processed by the Backend Framework, which handles data validation, preprocessing, and interaction with the machine learning components. The processed data is passed to the Ensemble Prediction Engine, where multiple models like KNN, SVM, and Random Forest analyze the input. These models work together using a Voting or Stacking classifier, which combines their predictions to produce a more accurate and reliable result. The final prediction is then stored in the Prediction Results Database for future reference and analysis.

The result is sent back to the user interface as an API response, where it is displayed in a simple and understandable format. Additionally, the stored data can be accessed through an Admin Dashboard to view reports and monitor system performance.

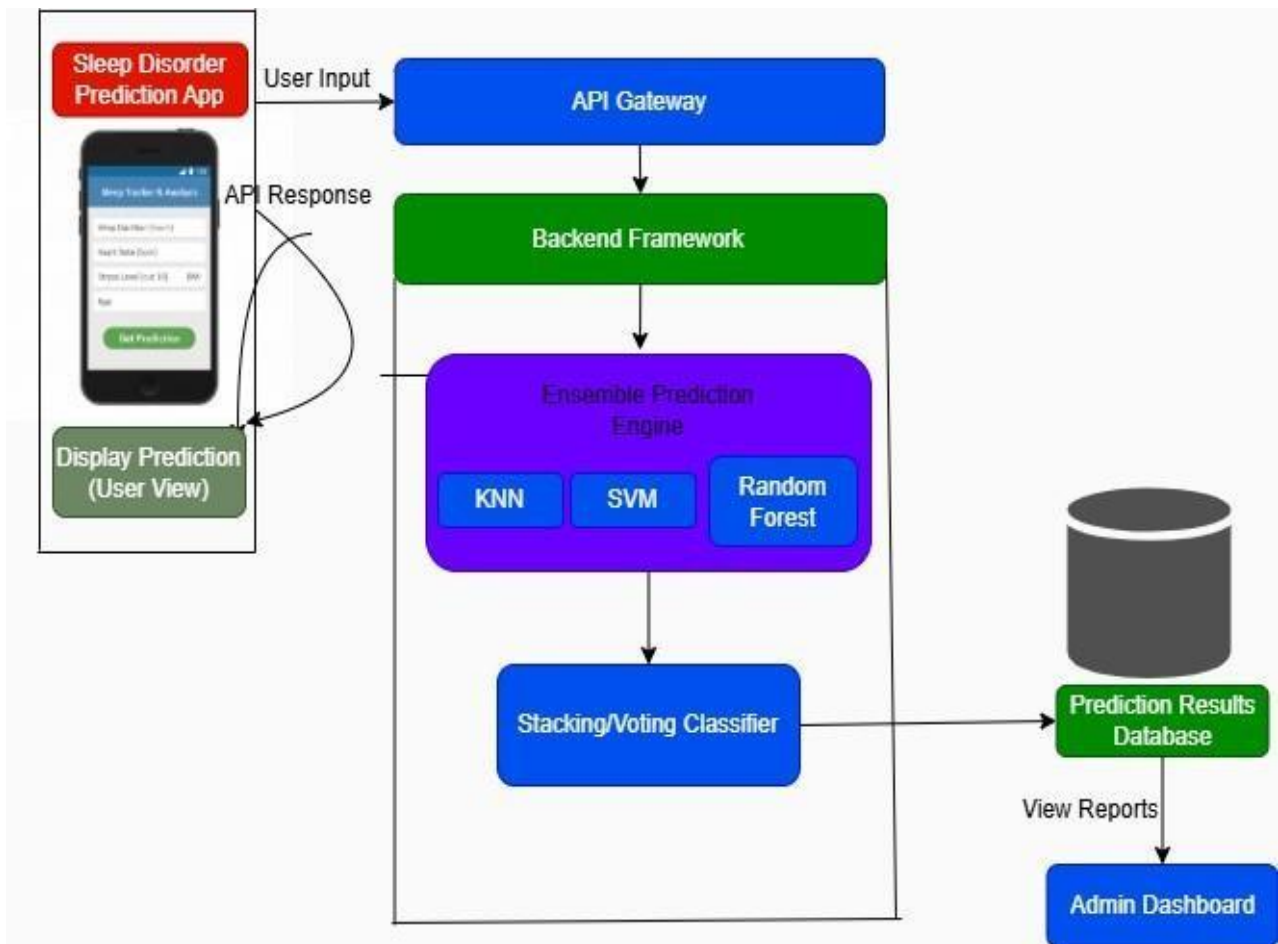


Fig 3.1. Architecture

3.3 Algorithms

3.3.1 K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a simple, non-parametric algorithm used for classification and regression tasks. It classifies a data point based on the majority class among its k-nearest neighbors, where k is a user-defined parameter. The algorithm computes the distance between the data point and all points in the training set using distance metrics such as Euclidean, Manhattan, or Minkowski. The k points that are closest to the data point are identified, and the most common class among these neighbors determines the classification result.

In this project, KNN is applied to classify sleep disorders using health and lifestyle data. First, the dataset is preprocessed by normalizing numerical features (e.g., age, sleep duration, physical activity level) and encoding categorical variables (e.g., gender, occupation). Missing values are handled through imputation or removal. The data is then split into training and testing sets. The KNN algorithm is trained on the training set, where it learns to classify data points based on their nearest neighbors. During prediction, the algorithm calculates the distance between a new data point and all training points, identifies the k-nearest neighbors, and assigns the class based on the majority vote among these neighbors. The value of k is optimized using cross-validation to ensure the best performance. The performance of KNN is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrices. While KNN is easy to implement and understand, it can be computationally expensive with large datasets, and its performance can be affected by the choice of k and distance metric. In the context of this project, KNN provides a baseline for comparison with more advanced algorithms.

3.3.2 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a powerful and versatile supervised learning algorithm used for classification and regression tasks. It works by finding the optimal hyperplane that best separates the data points of different classes in a high-dimensional space. The hyperplane is chosen to maximize the margin, which is the distance between the hyperplane and the nearest data points from each class, known as support vectors. SVM can handle linear and non-linear classification through the use of kernel functions, which transform the input data into a higher-dimensional space where it becomes linearly separable.

The preprocessing steps involve normalizing numerical features, encoding categorical variables, and handling missing values. The preprocessed data is then divided into training and testing sets. The SVM algorithm is trained on the training set, where it learns to identify the optimal hyperplane that separates the data points of different classes. The choice of kernel function (e.g., linear, polynomial, radial basis function) is crucial and is determined through cross-validation. During prediction, SVM projects new data points into the high-dimensional space and determines their class based on the side of the hyperplane they fall on. The performance of SVM is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrices. SVM is effective in high-dimensional spaces and works well with clear margin separation between classes. However, it can be computationally intensive and sensitive to the choice of hyperparameters such as the regularization parameter (C) and kernel parameters. In the context of this project, SVM provides a robust method for classifying sleep disorders, particularly when the data is not linearly separable.

3.3.3 Random Forest

Random Forest is an ensemble learning algorithm that builds multiple decision trees and merges their predictions to improve accuracy and control overfitting. Each tree is trained on a bootstrap sample of the data, and at each split, a random subset of features is considered. The final prediction is made by averaging the predictions of all trees (for regression) or by majority voting (for classification). This randomization helps in creating a diverse set of trees, which collectively produce a more robust and accurate model.

In this project, Random Forest is utilized to classify sleep disorders using health and lifestyle data. The preprocessing steps involve normalizing numerical features, encoding categorical variables, and handling missing values. The preprocessed data is split into training and testing sets. The Random Forest algorithm is trained on the training set, where multiple decision trees are built using different bootstrap samples and feature subsets. Each tree independently classifies the data, and the final classification is determined by aggregating the predictions of all trees. The performance of the Random Forest is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrices. Random Forest is advantageous because it reduces the risk of overfitting and improves predictive performance by averaging multiple decision trees. It is also less sensitive to the choice of hyperparameters compared to individual decision trees. However, it can be computationally intensive and requires careful tuning of parameters such as the number of trees and the maximum depth of each tree. In the context of this project, Random Forest serves as a

powerful ensemble method for classifying sleep disorders and provides a robust baseline for comparison with more advanced techniques.

3.3.4 Voting Classifier

A Voting Classifier is an ensemble learning technique that combines the predictions of multiple machine learning models to generate a single, more accurate and reliable output. Instead of depending on a single algorithm, this method integrates models such as KNN, SVM, and Random Forest, where each model independently analyzes the input data and produces its own prediction. The final decision is made by aggregating these individual predictions using a voting strategy. In hard voting, the class that receives the majority of votes from the models is selected as the final output, whereas in soft voting, the probabilities predicted by each model are averaged, and the class with the highest overall probability is chosen. This approach is particularly useful because different models have different strengths and weaknesses; for example, one model may perform well with certain patterns in the data while another may handle noise or non-linear relationships better. By combining them, the Voting Classifier reduces the chances of incorrect predictions made by any single model. It also helps in minimizing overfitting and improves the generalization ability of the system when applied to new, unseen data.

In the proposed sleep disorder prediction system, the Voting Classifier plays a crucial role in enhancing prediction accuracy by utilizing real-time user input such as sleep duration, heart rate, stress level, BMI, and age. Each model processes these inputs and contributes to the final decision, ensuring that the prediction is not biased toward a single algorithm. Additionally, the Voting Classifier provides consistency and robustness in results, making the system more dependable for users. Overall, it acts as a powerful technique to improve performance, stability, and reliability in machine learning-based healthcare applications.

3.4 Module Description

System Workflow:

1. Collect Data:

Collect the data of health and lifestyle data relevant to sleep disorders. This data should include various features such as age, gender, occupation, sleep duration, quality of sleep, physical

activities, BMI category, blood pressure, heart rate, snoring frequency, working hours.

2. Data Preprocessing:

Once the data is loaded, undergo data cleaning and preprocessing procedures. This involves handling missing or corrupted data, encoding categorical variables, normalizing or standardizing numerical features, and applying data augmentation techniques if necessary to improve model generalization.

3. Model Building:

Design and implement suitable ensemble learning architectures, such as Stacking or Voting Classifiers, for the classification task. Train the models using the preprocessed data, tuning hyperparameters to optimize performance. The base models for stacking might include K-Nearest Neighbors (KNN), Support Vector Machine (SVM) and Random Forest as the meta-model.

4. Model Prediction:

Use the trained ensemble models to generate predictions on new, unseen health and lifestyle data. This involves preprocessing the new data similarly and using the models to predict the presence or absence of sleep disorders.

5. Result:

Present the results, including the predicted sleep disorder status for each data instance.

User Interaction:

1.Register:

Users should first register with their credentials to create an account in the system.

2.Login:

Users can log in with their registered credentials to access the system.

3.Enter Input Data:

Users can enter their health and lifestyle data, including various features such as age, gender, occupation, sleep duration, quality of sleep, physical activity level, stress level, BMI category, blood pressure, heart rate,

and Working hours. This data should be in a structured format compatible with the system.

4. Viewing Results:

After enter their data, users can view the classification results provided by the model. The system will display the predicted sleep disorder status along with advisory and health recommendations.

5. Logout:

Finally, users can log out of the system to secure their session and personal data.

3.5 UML Diagrams

- UML, an acronym for Unified Modeling Language, is a versatile modeling language extensively applied in the realm of object-oriented software engineering. Oversight and development of its standards are managed by the Object Management Group.
- The primary objective of UML is to establish a standardized language for modeling computer programs rooted in object-oriented principles. Its components currently encompass a meta-model and a notation, with potential for future augmentation through methods or processes.
- Beyond software engineering, the Unified Modeling Language finds utility as a standardized means for business modeling and describing, visualizing, constructing, and documenting software system components. Its foundations stem from a compilation of effective engineering principles suitable for complex system simulation.
- Vital to the software development process and the realization of object-oriented software, UML serves as an integral element. It predominantly employs visual notations to convey the design aspects of software projects.

Goals

The key objectives underlying the creation of the UML encompass:

1. Empowering users with a readily accessible and expressive visual modeling language, facilitating the development and interchange of substantial models.
1. Establishing mechanisms for extendibility and specialization, enabling the augmentation of core concepts.
2. Ensuring autonomy from specific programming languages and development methodologies.
3. Erecting a formal foundation for comprehending the modeling language.
4. Fostering the expansion of the market for object-oriented tools.
5. Facilitating the incorporation of advanced development notions such as collaborations, frameworks, patterns, and components.
6. Seamlessly integrating exemplary practices.
7. Improving Communication Among Stakeholders
8. Providing Standardized Notation
9. It offers a common and standardized way to represent system design, ensuring consistency across different teams and projects.
10. Supporting System Visualization at Multiple Levels

3.5.1 Class Diagram

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

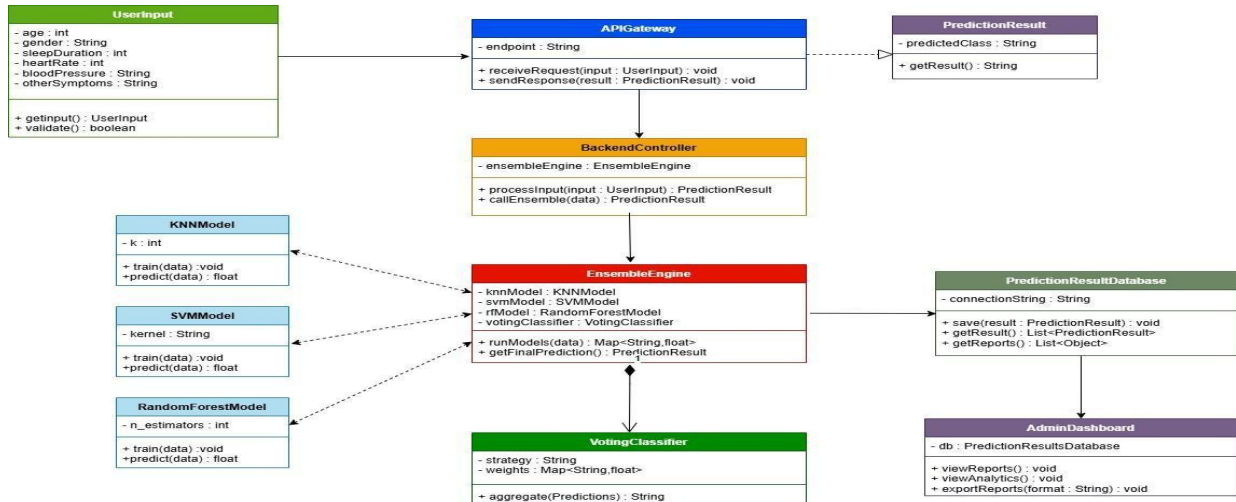


Fig 3.2 Class Diagram

3.5.2 Use Case Diagram

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show



Fig 3.3 Use Case Diagram

3.5.3 Sequence Diagram

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams. It focuses on the flow of messages exchanged between objects (like user interface, backend, and machine learning model) to complete a specific task. In your Sleep Disorder Prediction System, it explains how the user request moves through the system and how the response is generated.

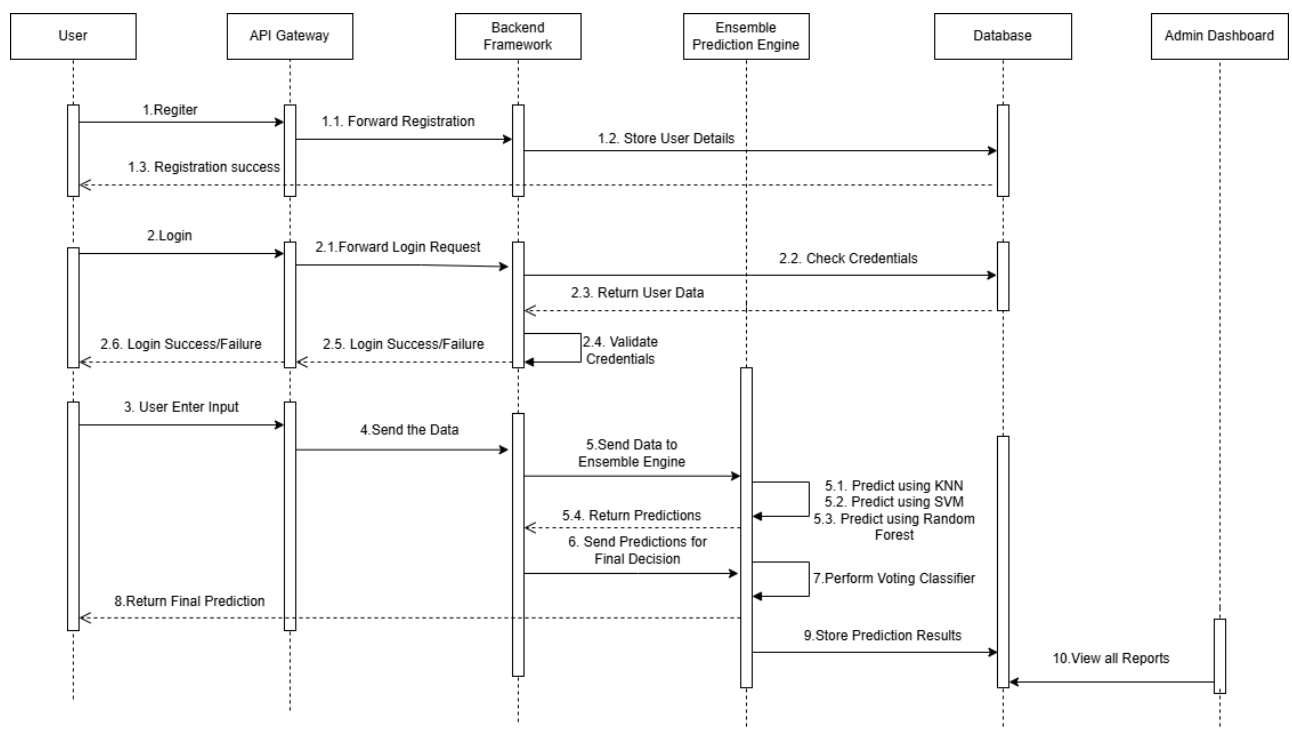


Fig 3.4 Sequence Diagram

3.5.4 Activity Diagram

An activity diagram comprehensively illustrates the entire flow of control. Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. Activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

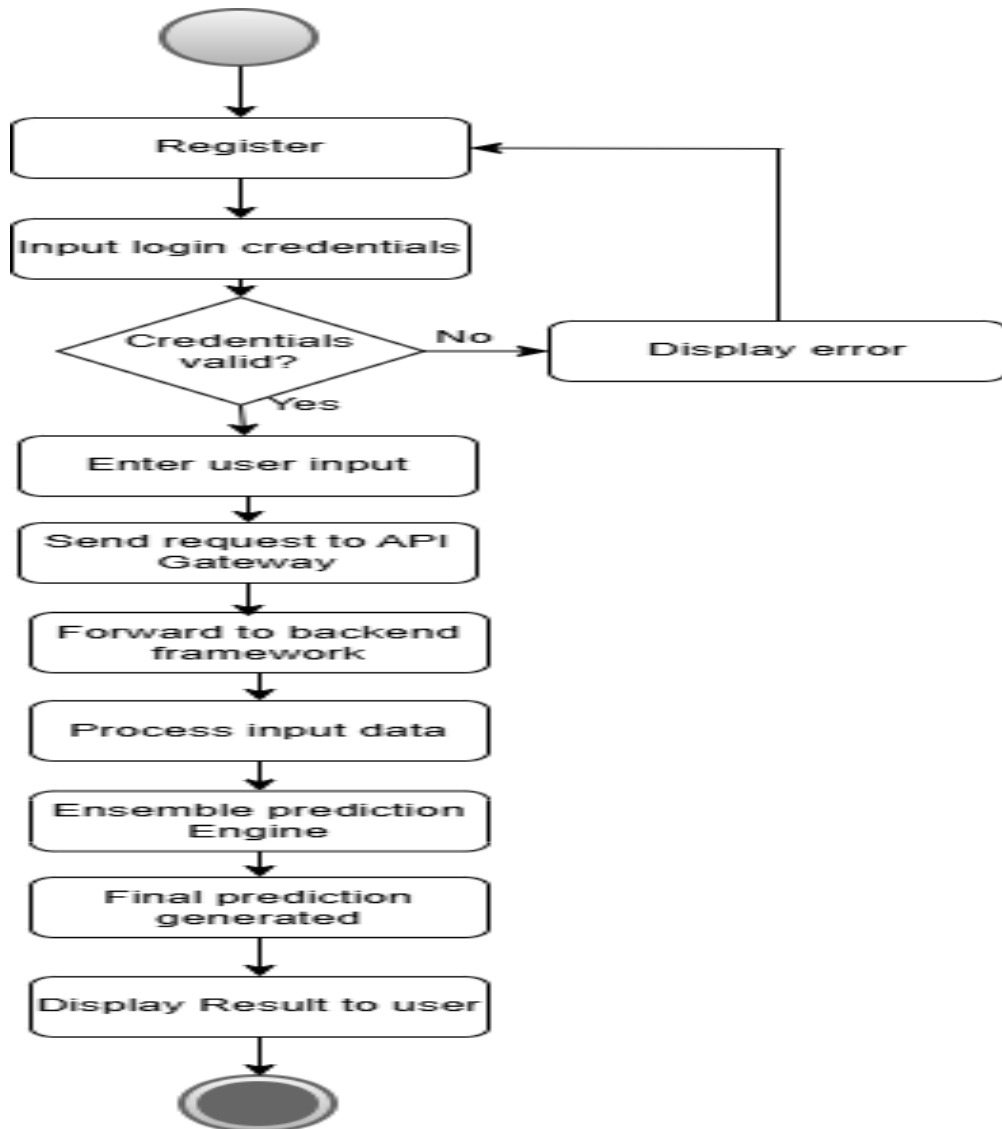


Fig 3.5 Activity Diagram

3.5.5 Collaboration Diagram

In a collaboration diagram, the sequence of method calls is represented through a numbering method, as depicted in the example below. This numbering reflects the sequence of method invocations. The same order management system is used to elucidate the collaboration diagram. While the method calls bear resemblance to those in a sequence diagram, the collaboration diagram showcases the arrangement of objects, in contrast to the sequence diagram that solely outlines the sequence.

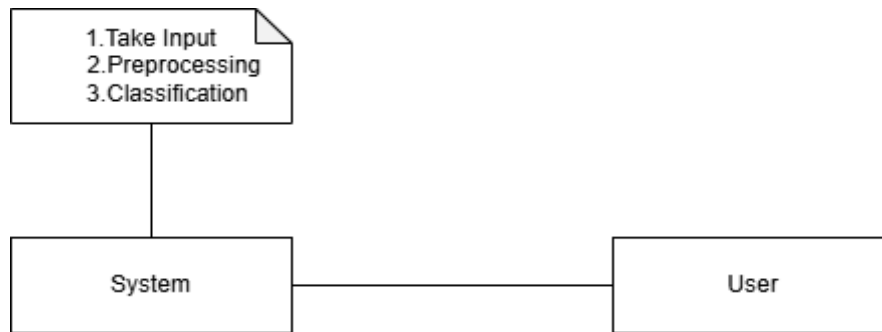


Fig 3.6 Collaboration Diagram

3.5.6 Deployment Diagram

A depiction of the system's deployment perspective is realized through a deployment diagram, which maintains a connection with the component diagram. This connection arises from the fact that deployment diagrams facilitate the deployment of components. Within a deployment diagram, nodes play a central role, representing the tangible hardware elements responsible for executing the software application.

Fig 3.7 Deployment Diagram

3.5.7 Component Diagram

Referred to as a UML component diagram, a component diagram offers a visual representation of the arrangement and structural organization of a system's tangible components. These diagrams are commonly crafted to elucidate implementation specifics and ensure that the system's intended development comprehensively meets its requirements.

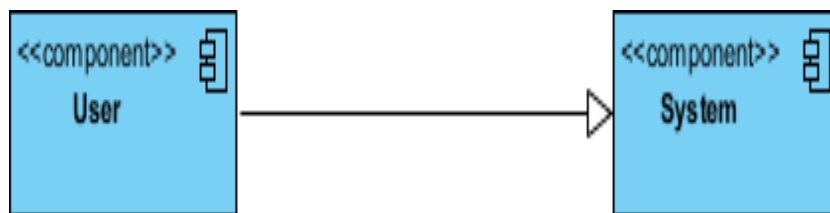


Fig 3.8 Component Diagram

3.5.8 ER Diagram

Utilizing an entity-relationship model (ER model), the structure of a database is visually represented through an entity relationship diagram (ER Diagram). This diagram serves as a blueprint or design for creating a future database. The foundational components of the E-R model consist of the entity set and the relationship set.

An ER diagram elucidates the associations among various entity sets. An entity set comprises interconnected entities, each potentially possessing distinct properties. In a database management system, an entity corresponds to a table or an attribute within a table. Consequently, the ER diagram effectively portrays the comprehensive logical organization of a database by illustrating the interconnections between tables and their corresponding attributes. Let's consider a simple example of an ER diagram.

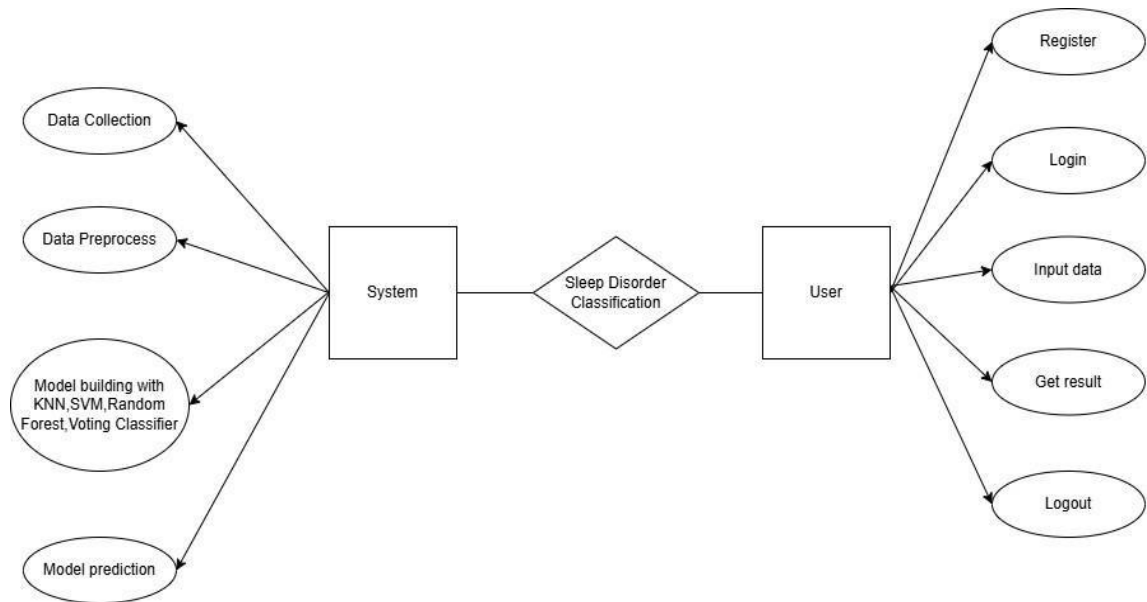


Fig 3.9 ER Diagram

3.5.9 DFD Diagram

Historically, the depiction of internal information flows within a system has been achieved through the utilization of data flow diagrams (DFDs). A well-structured and comprehensible DFD has the capacity to visually encapsulate a significant portion of the system's requirements.

This can be accomplished manually, through automation, or through a combination of both techniques.

The DFD showcases the trajectory of data as it enters and exits the system, discerns points of data alteration, and identifies storage locations. The overarching purpose of a DFD is to delineate the scope and boundaries of an entire system. It can effectively serve as a means of communication between a systems analyst and all stakeholders involved in the system, offering a foundational tool for the process of system redesign.

3.5.9.1 Context Diagram:

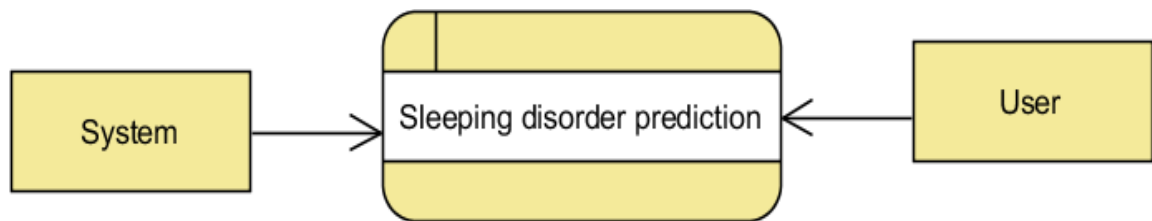


Fig 3.10: Context Diagram

3.5.9.2 DFD Level-1 Diagram:

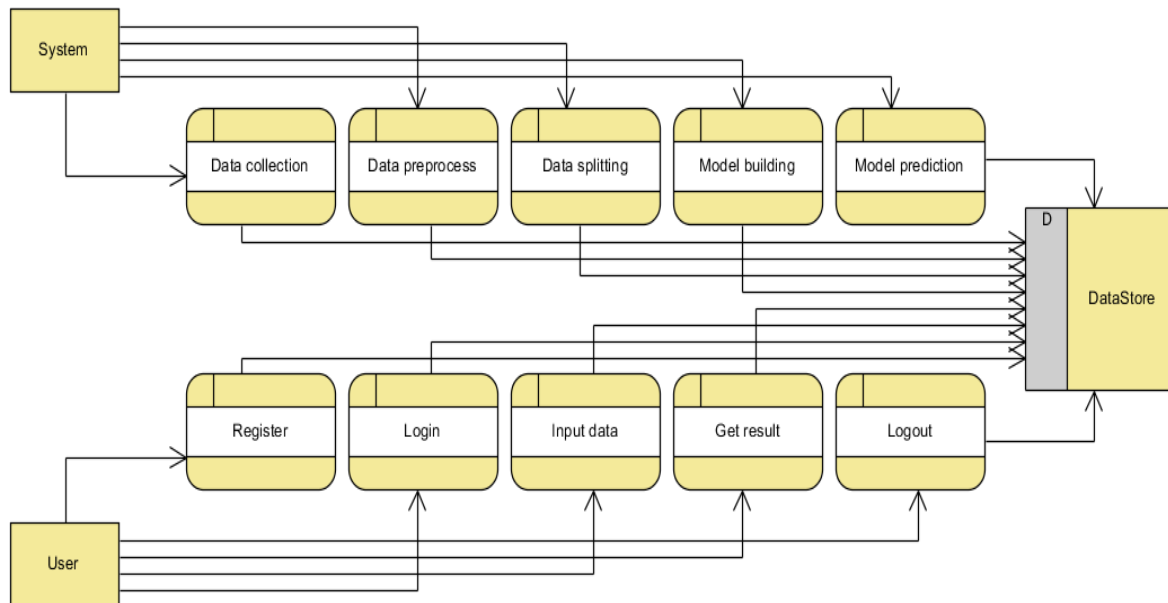


Fig 3.11 DFD Level-1 Diagram

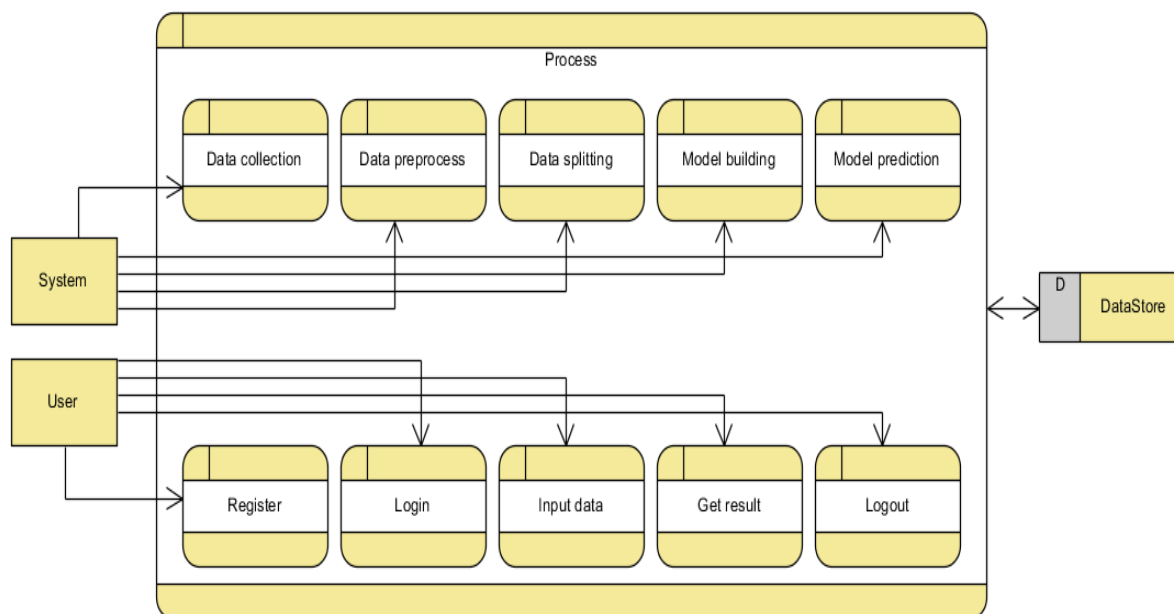
3.5.9.3. DFD Level-2 Diagram:

Fig 3.12 DFD Level-2 Diagram

CHAPTER 4

PROPOSED SYSTEM AND ANALYSIS

4.1. Introduction

The proposed system, titled “An Ensemble Learning Approach for Improved Sleep Disorder Prediction,” is designed to provide an efficient, accurate, and user-friendly solution for identifying sleep disorders using machine learning techniques. Sleep disorders such as insomnia, sleep apnea, and restless sleep have become increasingly common due to modern lifestyle factors, stress, and irregular sleep patterns. Traditional diagnostic methods like Polysomnography (PSG), although accurate, are expensive, time-consuming, and require clinical supervision, making them less accessible for regular use.

To address these challenges, the proposed system introduces a machine learning-based prediction model that utilizes an ensemble learning approach. Instead of relying on a single algorithm, the system combines multiple models such as K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest. These models work together using a Voting classifier to generate a final prediction, improving overall accuracy, consistency, and robustness. The ensemble approach helps overcome the limitations of individual models by leveraging their strengths and reducing prediction errors.

The system is designed to work with real-time user input data, allowing users to enter parameters such as sleep duration, heart rate, stress level, BMI, and age through a web-based interface. This makes the system more practical and accessible compared to existing solutions that depend on pre-collected datasets or clinical equipment. The backend processes the input data, performs necessary preprocessing, and feeds it into the ensemble model for prediction. The result is then displayed to the user in a clear and understandable format.

From an analytical perspective, the proposed system focuses on improving prediction accuracy, usability, and scalability. The use of ensemble learning significantly enhances

performance metrics such as accuracy, precision, recall, and F1-score when compared to individual models. Additionally, the system reduces overfitting and increases generalization capability, making it more reliable for real-world applications. The integration of a web-based platform ensures ease of use, while the modular architecture supports future scalability and enhancements.

Furthermore, the system addresses key gaps identified in existing research, such as lack of real-time prediction, limited feature consideration, and absence of user-friendly interfaces. By combining advanced machine learning techniques with practical implementation, the proposed system provides a cost-effective and accessible solution for early detection of sleep disorders. This can help users take preventive measures and improve their overall health and well-being.

4.2. Requirement Analysis

Requirement Analysis is the process of identifying, analyzing, and defining the needs and functionalities required to develop the proposed system. It helps in understanding what the system should do and how it should perform. In this project, requirement analysis focuses on both functional and non-functional aspects to ensure an efficient and reliable sleep disorder prediction system.

4.2.1 Functional Requirements

Functional requirements describe the core operations of the system.

- The system should allow users to **enter health-related input data** such as sleep duration, heart rate, stress level, BMI, and age.
- The system should **validate user input** to ensure accuracy and completeness.
- The system should **process input data** using machine learning models (KNN, SVM, Random Forest).
- The system should implement an **ensemble learning technique (Voting Classifier)** to generate the final prediction.
- The system should **predict the presence or absence of sleep disorders** based on input data.

- The system should **display results** clearly to the user.
- The system should **store prediction results** in a database for future reference.
- The system should provide an **admin interface/dashboard** to monitor and manage data.
- Functional encompass the specific specifications that a system must adhere to in order to fulfill the essential needs of its end users. These features are mandatory components of the system as per the contractual obligations. They are articulated as the input data required, the actions to be executed, and the intended outcomes. In contrast to non-functional requirements, functional requirements are essentially the user-defined criteria that manifest visibly in the final product.
- **For instance:**
 - User authentication upon logging into the system.
 - Automatic system shutdown in response to a cyber-attack.
 - Sending a verification email to a user upon their initial registration within a software system.

4.2.2 Non-Functional Requirements

- Non-functional Requirements encapsulate the quality prerequisites stipulated by the project agreement that the system must fulfill. The significance and application level of these criteria can vary across different projects. They are also referred to as non-behavioral requirements. These requirements address concerns such as:
 - Portability
 - Security
 - Maintainability
 - Reliability
 - Scalability
 - Performance
 - Reusability
 - Flexibility
 - Examples of non-functional requirements are:
 - Emails must be dispatched with a latency not exceeding 12 hours from a

specific activity.

- Each request processing should be completed within a 10- second timeframe.

4.2.3 System Requirements

System requirements define the technical specifications and resources needed for the successful development, deployment, and execution of the proposed **Sleep Disorder Prediction System**. These requirements ensure that the system performs efficiently, reliably, and accurately while handling real-time user inputs and machine learning predictions.

4.2.3.1 Hardware Requirements

The hardware requirements specify the physical components necessary to run the system smoothly:

Processor: Intel i3 or higher (i5 recommended for better performance)

RAM: Minimum 4 GB (8 GB recommended for handling machine learning operations efficiently)

Storage: At least 500 GB hard disk or SSD for faster data processing

Input Devices: Keyboard and mouse for user interaction

Display: Standard monitor

Internet Connection: Required for web-based access and data communication

4.2.3.2 Software Requirements

The software requirements include the tools, technologies, and platforms used to develop and run the system:

- **Operating System:** Windows
- **Programming Language:** Python
- **Frontend Technologies:** HTML, CSS, JavaScript
- **Backend Framework:** Flask
- **Machine Learning Libraries:** Scikit-learn, Pandas, NumPy
- **Database:** JSON
- **Development Tools:** VS Code

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Implementation

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Classification of Sleep Disorders</title>
  <style>
    * { margin: 0; padding: 0; box-sizing: border-box; }
    body { font-family: 'Arial', sans-serif; background-color: #050a14; color: white; line-
      height: 1.6; }

    nav {
      position: fixed; top: 0; width: 100%; display: flex;
      justify-content: space-between; align-items: center;
      padding: 20px 80px; background: rgba(5, 10, 20, 0.95); z-index: 1000;
    }
    .logo { font-size: 1.2rem; font-weight: bold; text-transform: uppercase; }
    .logo span { color: #f39c12; }
    .nav-links a { color: white; text-decoration: none; margin-left: 30px; font-weight: bold;
font-size: 0.85rem; }
    .admin-btn { border: 2px solid #f39c12; padding: 5px 15px; border-radius: 5px; color:
#f39c12 !important; }

    .hero {
      height: 100vh; background: linear-gradient(rgba(0,0,0,0.7), rgba(0,0,0,0.7)),
```

```
url('https://images.unsplash.com/photo-1541781774459-  
background-size: cover; display: flex; flex-direction: column; justify-content: center;  
align-items: center; text-align: center;  
}  
.hero h1 { font-size: 3rem; max-width: 800px; text-transform: uppercase; padding: 0  
20px; }  
.hero span { color: #f39c12; }  
  
.about-section { padding: 100px 80px; background: #0b1120; display: flex; align-items:  
center; gap: 50px; justify-content: center; }  
.about-image { flex: 0 0 380px; }  
.about-image img { width: 100%; border-radius: 8px; box-shadow: 0 10px 30px  
rgba(0,0,0,0.5); }  
.about-text { flex: 1; max-width: 600px; }  
.about-text h2 { font-size: 2.5rem; margin-bottom: 20px; color: #f39c12; }  
/* Modal Styles */  
.modal {  
display: none; position: fixed; top: 50%; left: 50%; transform: translate(-50%, -50%);  
background: white; color: #333; padding: 40px; border-radius: 10px; z-index: 2000;  
width: 400px;  
box-shadow: 0 10px 40px rgba(0,0,0,0.8);  
}  
.modal h2 { margin-bottom: 20px; text-align: center; color: #050a14; }  
.modal input { width: 100%; padding: 12px; margin: 10px 0; border: 1px solid #ccc;  
border-radius: 5px; display: block; }  
.submit-btn { background: #f39c12; color: white; width: 100%; padding: 12px; border:  
none; font-weight: bold; cursor: pointer; margin-top: 10px; border-radius: 5px; }  
.submit-btn:hover { background: #e67e22; }  
.overlay { display: none; position: fixed; top: 0; left: 0; width: 100%; height: 100%;  
background: rgba(0,0,0,0.85); z-index: 1500; }  
.switch-text { text-align: center; margin-top: 15px; font-size: 0.9rem; }
```

```
.switch-text a { color: #3498db; text-decoration: none; }

</style>
</head>

<body>
  <nav>
    <div class="logo">CLASSIFICATION OF <span>SLEEP DISORDERS</span></div>
    <div class="nav-links">
      <a href="/">HOME</a>
      <a href="#about">ABOUT</a>
      <a href="javascript:void(0)" onclick="openModal('regModal')">REGISTER</a>
      <a href="javascript:void(0)" onclick="openModal('loginModal')">LOGIN</a>
      <a href="/admin-login" class="admin-btn">ADMIN</a>
    </div>
  </nav>

  <section class="hero">
    <h1>AN ENSEMBLE LEARNING APPROACH FOR IMPROVED <span> SLEEP
DISORDER PREDICTION</span></h1>
  </section>

  <section id="about" class="about-section">
    <div class="about-image">
      <imgsrc="https://images.unsplash.com/photo-1551076805-
e1869033e561?q=80&w=2070&auto=format&fit=crop" alt="Sleep Study">
    </div>
    <div class="about-text">
      <h2>About</h2>
      <div style="margin-bottom: 60px; color: #d1d1d1; font-size: 1.05rem; text-align:
justify;">
        <p style="margin-top: 10px; font-size: 0.95rem; color: #ccc;">
          This project is a <strong>web-based sleep disorder prediction system</strong> designed to
```


help users assess their sleep health using personal and lifestyle data. The application collects details such as age, gender, blood pressure, heart rate, sleep duration, BMI category, snoring pattern, diet habits, stress-related behavior, previous health history, and work hours to generate a prediction report. It uses an **ensemble learning approach** that combines **K-Nearest**

Neighbors (KNN), **Support Vector Machine (SVM)**, and **Random Forest** models to classify the user's condition into

Low, **Moderate**, or **High** risk levels. In addition to prediction, the system provides secure user registration and login, stores generated reports for later review, and includes an admin module for monitoring records and supporting doctor consultation workflow.

```
</p>
</div>
</div>
</section>
<div id="overlay" class="overlay" onclick="closeModals()"></div>
<div id="regModal" class="modal">
  <h2>Register</h2>
  <form id="registrationForm" action="/register" method="POST">
    <input type="text" name="name" placeholder="Full Name" required>
    <div style="display: flex; gap: 5px; margin-bottom: 10px;">
      <input type="email" id="regEmail" name="email" placeholder="Email Address"
required style="flex: 1; margin: 0;">
      <button type="button" onclick="sendOTP('email')" style="background:#3498db;
color:white; border:none; padding:0 10px; border-radius:5px; cursor:pointer; font-
size:0.8rem;">Send OTP</button>
    </div>
    <input type="tel" id="regPhone" name="phone" placeholder="Phone Number
(Mandatory for Doctor Consultation)" pattern="[6-9][0-9]{9}" required style="width:100%;
margin: 0 0 10px 0; padding:12px; border:1px solid #ccc; border-radius:5px; display:block;">
```

```
<input type="password" name="password" placeholder="Password (Min 8 chars)"
required>
<div id="otpInputSection" style="display:none; margin-top: 15px; border: 1px dashed
#f39c12; padding: 15px; border-radius: 8px; background: #fdfaf4;">
  <p style="font-size: 0.85rem; color: #333; margin-bottom: 10px;">Enter the 6-digit OTP
sent to your email:</p>
  <input type="text" id="otpValue" placeholder="Enter OTP" style="margin-bottom:
10px;">
  <button type="button" onclick="verifyOTP()" class="submit-btn" style="background:
#2ecc71; margin: 0;">VERIFY & REGISTER</button>

</div>
<button      type="submit"      id="mainSubmitBtn"      class="submit-btn">CREATE
ACCOUNT</button>
</form>
  <div class="switch-text">Already have an account? <a href="javascript:void(0)"
onclick="switchModal('loginModal')">Login</a></div>
</div>
<div id="loginModal" class="modal">
  <h2>Login</h2>
  <form action="/login" method="POST">
    <input type="email" name="email" placeholder="Email Address" required>
    <input type="password" name="password" placeholder="Password" required>
    <button type="submit" class="submit-btn">LOG IN</button>
  </form>
  <div class="switch-text">New user?<a href="javascript:void(0)":
onclick="switchModal('regModel')>Register</a><a href="javascript:void(0)"
onclick="switchModal('forgotPasswordModel')>Forgot Password?</a></div>
</div>
<div id="forgotPasswordModal" class="modal">
  <h2>Reset Password</h2>
  <form id="forgotPasswordForm">
```

```
<input type="email" id="forgotEmail" placeholder="Enter your Email Address"
required>
<button type="button" onclick="sendResetOTP()" class="submit-btn"
style="background: #3498db;">Send Reset OTP</button>
</form>
<div id="resetOtpSection" style="display:none; margin-top: 15px;">
<input type="text" id="resetOtpValue" placeholder="Enter OTP" style="width:100%;
padding:12px; border:1px solid #ccc; border-radius:5px; margin-bottom:10px;
display:block;">
<input type="password" id="newPasswordValue" placeholder="New Password (Min 8
chars)" style="width:100%; padding:12px; border:1px solid #ccc; border-radius:5px; margin-
bottom:10px; display:block;">
<button type="button" onclick="resetPassword()" class="submit-btn"
style="background: #2ecc71;">RESET PASSWORD</button>

</div>
<script>
function openModal(id) {
document.getElementById(id).style.display = 'block';
document.getElementById('overlay').style.display = 'block';
}

function closeModals() {
document.getElementById('regModal').style.display = 'none';
document.getElementById('loginModal').style.display = 'none';
document.getElementById('forgotPasswordModal').style.display = 'none';
document.getElementById('resetOtpSection').style.display = 'none';
document.getElementById('overlay').style.display = 'none';
// Reset form fields
document.getElementById('forgotEmail').value = "";
document.getElementById('resetOtpValue').value = "";
document.getElementById('newPasswordValue').value = "";
```

```
    }

    function switchModal(id) {
        closeModals();
        setTimeout(() => openModal(id), 100);
    }

    function sendOTP(type) {
        const value = document.getElementById('regEmail').value;
        if(!value) { alert("Please enter email first!"); return; }
        fetch('/send-otp', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ type: 'email', receiver: value })

        })

        .then(res => {
            if(res.ok) {
                document.getElementById('otpInputSection').style.display = 'block';
                document.getElementById('mainSubmitBtn').style.display = 'none';
                alert("OTP has been sent to your email");
            } else {
                alert("Error sending OTP!");
            }
        });
    }

    function verifyOTP() {
        const otp = document.getElementById('otpValue').value;
        fetch('/verify-otp', {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
```

```
        body: JSON.stringify({ otp: otp })
      })
      .then(res => res.json())
      .then(data => {
        if(data.success) {
          alert("Email verified successfully!");
          document.getElementById('registrationForm').submit();
        } else {
          alert("Invalid OTP!");
        }
      });
    });

function sendResetOTP() {
  const email = document.getElementById('forgotEmail').value;

  if(!email) { alert("Please enter email address!"); return; }

  fetch('/forgot-password', {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({ email: email })
  })
  .then(res => res.json())
  .then(data => {
    if(data.success) {
      document.getElementById('resetOtpSection').style.display = 'block';
      alert("OTP sent to your email address");
    } else {
      alert(data.message || "Email not found");
    }
  })
}
```

```
    });  
  }  
  function resetPassword() {  
    const email = document.getElementById('forgotEmail').value;  
    const otp = document.getElementById('resetOtpValue').value;  
    const newPassword = document.getElementById('newPasswordValue').value;  
  
    if(!email || !otp || !newPassword) { alert("Please fill all fields!"); return; }  
  
    fetch('/reset-password', {  
      method: 'POST',  
      headers: { 'Content-Type': 'application/json' },  
      body: JSON.stringify({ email: email, otp: otp, newPassword: newPassword })  
    })  
  
    .then(res => res.json())  
      .then(data => {  
  
        if(data.success) {  
          alert("Password reset successfully!");  
          closeModals();  
          setTimeout(() => openModal('loginModal'), 100);  
        } else {  
          alert(data.message || "Error resetting password");  
        }  
      });  
    }  
  </script>  
</body>  
</html>
```

Server.js:

```
const express = require('express');
const bcrypt = require('bcryptjs');
const bodyParser = require('body-parser');
const path = require('path');
const fs = require('fs');
const KNN = require('ml-knn');
const { RandomForestClassifier } = require('ml-random-forest');
const SVM = require('libsvm-js/asm');
const app = express();
app.use(express.json());
app.use(express.urlencoded({ extended: true }));
app.use(bodyParser.urlencoded({ extended: true }));
// This line allows the server to serve your index.html correctly
app.use(express.static(_dirname));
const session = require('express-session');
app.use(session({
  secret: 'sleep-disorder-admin-secret-2025',
  resave: false,

  saveUninitialized: false,
  cookie: { maxAge: 3600000 }
}));
const USERS_DB = './database.json';
const REPORTS_DB = './reports.json';
const ADMIN_DB = './admin_database.json';
const CALL_LOGS_DB = './admin_call_logs.json';
const asteriskAriUrl = process.env.ASTERISK_ARI_URL || "";
const asteriskAriUser = process.env.ASTERISK_ARI_USERNAME || "";
const asteriskAriPass = process.env.ASTERISK_ARI_PASSWORD || "";
const asteriskAriApp = process.env.ASTERISK_ARI_APP || 'sleep-disorder-app';
```

```
const asteriskEndpointPrefix = process.env.  
ASTERISK_ENDPOINT_PREFIX || "";  
const asteriskCallerId = process.env.ASTERISK_CALLER_ID || 'A1 Hospital';  
const hasAsteriskConfig = Boolean(  
  asteriskAriUrl &&  
  asteriskAriUser &&  
  asteriskAriPass &&  
  asteriskEndpointPrefix  
);  
if (!fs.existsSync(USERS_DB)) fs.writeFileSync(USERS_DB, JSON.stringify([]));  
if (!fs.existsSync(REPORTS_DB)) fs.writeFileSync(REPORTS_DB, JSON.stringify([]));  
if (!fs.existsSync(ADMIN_DB)) fs.writeFileSync(ADMIN_DB, JSON.stringify([]));  
if (!fs.existsSync(CALL_LOGS_DB)) fs.writeFileSync(CALL_LOGS_DB,  
JSON.stringify([]));  
const readData = (file) => JSON.parse(fs.readFileSync(file, 'utf8'));  
const writeData = (file, data) => fs.writeFileSync(file, JSON.stringify(data, null, 2));  
const MAX_WEEKLY_ENTRIES = 3;  
const WEEK_IN_MS = 7 * 24 * 60 * 60 * 1000;  
function getReportTimestamp(report) {  
  if (report.createdAt) {  
    const createdAtMs = Date.parse(report.createdAt);  
    if (Number.isFinite(createdAtMs)) return createdAtMs;  
  }  
  const idMs = Number(report.id);  
  if (Number.isFinite(idMs)) return idMs;  
  const dateMs = Date.parse(report.date);  
  if (Number.isFinite(dateMs)) return dateMs;  
  return 0;  
}  
function countRecentUserReports(reports, email, nowMs = Date.now()) {  
  const cutoff = nowMs - WEEK_IN_MS;
```



```
    return reports.filter((report) => report.email === email && getReportTimestamp(report) >=
cutoff).length;
}

function countRecentPhoneReports(reports, phone, nowMs = Date.now()) {
    const cutoff = nowMs - WEEK_IN_MS;
    const canonical = toCanonicalIndianPhone(phone);
    return reports.filter((report) => toCanonicalIndianPhone(report.phone_number || "") ===
canonical && getReportTimestamp(report) >= cutoff).length;
}

function isValidIndianPhoneNumber(phone = "") {
    const digits = String(phone || "").replace(/^\D/g, "").trim();
    if (/^91[6-9]\d{9}$/.test(digits)) return true;
    return /^[6-9]\d{9}$/.test(digits);
}

function toCanonicalIndianPhone(phone = "") {
    const digits = String(phone || "").replace(/^\D/g, "").trim();
    if (/^91[6-9]\d{9}$/.test(digits)) return digits.slice(2);
    if (/^[6-9]\d{9}$/.test(digits)) return digits;
    return "";
}

function normalizePhoneForCloudCall(phone) {
    const canonical = toCanonicalIndianPhone(phone);
    if (!canonical) return "";

    return `+91${canonical}`;
}

function normalizePhoneForDisplay(phone) {
    return toCanonicalIndianPhone(phone);
}

function normalizePhoneForCallLink(phone) {
    const canonical = toCanonicalIndianPhone(phone);
```

```
    if (!canonical) return "";
    return `+91${canonical}`;
}
const RISK_LABELS = ['Low', 'Moderate', 'High'];
function clampNumber(value, min, max, fallback = 0) {
    const n = Number(value);
    if (!Number.isFinite(n)) return fallback;
    return Math.max(min, Math.min(max, n));
}
function parseBP(bpValue = "") {
    const parts = String(bpValue).split('/');
    const systolic = clampNumber(parts[0], 70, 220, 120);
    const diastolic = clampNumber(parts[1], 40, 140, 80);
    return { systolic, diastolic };
}
function encodeBmiCategory(bmi = "") {
    const value = String(bmi).toLowerCase();
    if (value.includes('underweight')) return 0;
    if (value.includes('normal')) return 1;
    if (value.includes('overweight')) return 2;
    if (value.includes('obese class i')) return 3;
    if (value.includes('obese class ii')) return 4;
    if (value.includes('obese class iii') || value.includes('extremely')) return 5;
    if (value.includes('obese')) return 4;
    return 1;
}
function encodeRiskLabel(record) {
    if (record.riskLevel && RISK_LABELS.includes(record.riskLevel)) {
        return RISK_LABELS.indexOf(record.riskLevel);
    }
    const resultText = String(record.result || "").toLowerCase();
```

```
    if (resultText.includes('high')) return 2;
    if (resultText.includes('moderate')) return 1;
    return 0;
  }
  function normalizeOccupation(occupation = "") {
    const raw = String(occupation || "").trim();
    if (!raw) return 'Professional';
    const cleaned = raw.replace(/[0-9]+/g, "").trim();
    if (!cleaned) return 'Professional';
    return cleaned.charAt(0).toUpperCase() + cleaned.slice(1).toLowerCase();
  }
  function toFeatureVector(record) {
    const { systolic, diastolic } = parseBP(record.bp);
    const genderMap = { male: 0, female: 1, other: 2 };
    const angerMap = { calm: 0, leave: 1, shouting: 2, beating: 3 };
    const dietMap = { daily: 0, sometimes: 1, never: 2 };
    const problemMap = { solve: 0, help: 1, ignore: 2 };
    const snoringMap = { never: 0, sometimes: 1, 'every night': 2 };
    const gender = String(record.gender || "").toLowerCase();
    const angerResponse = String(record.anger_response || "").toLowerCase();
    const diet = String(record.diet || "").toLowerCase();
    const problemSolving = String(record.problem_solving || "").toLowerCase();
    const snoring = String(record.snoring || "").toLowerCase();
    const prevHealth = String(record.prev_health || "").toLowerCase();
    return [
      clampNumber(record.age, 10, 100, 30),

      genderMap[gender] ?? 0,

      systolic,
      diastolic,
      clampNumber(record.heart_rate, 40, 200, 80),
```

```
    clampNumber(record.sleep_duration, 1, 14, 7),
    clampNumber(record.tea_coffee, 0, 15, 1),
    encodeBmiCategory(record.bmi),
    snoringMap[snoring] ?? 0,
    angerMap[angerResponse] ?? 0,
    dietMap[diet] ?? 1,
    problemMap[problemSolving] ?? 1,
    prevHealth === 'yes' ? 1 : 0,
    clampNumber(record.work_hours, 0, 120, 40)
  ];
}
function fallbackRiskLabel(data) {
  const sleep = Number(data.sleep_duration);
  const bmiCategory = data.bmi || "";
  if (bmiCategory.includes('Obese') && data.snoring === 'Every Night') return 2;
  if (sleep < 5) return 2;
  if (sleep < 6 && data.snoring === 'Every Night') return 1;
  if (data.diet === 'never' && data.problem_solving === 'ignore') return 2;

  if (data.diet === 'never' || data.problem_solving === 'ignore') return 1;
  return 0;
}
function buildTrainingSetFromReports(reports) {
  const X = [];
  const y = [];
  reports.forEach((row) => {
    X.push(toFeatureVector(row));
    y.push(encodeRiskLabel(row));
  });
  return { X, y };
```

```
app.listen(3000, '0.0.0.0', () => {  
    console.log('Server is running on http://localhost:3000');  
    console.log('Or on mobile at: http://YOUR_IP_ADDRESS:3000');  
});
```

ML_prediction.py:

```
import json  
import pandas as pd  
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier  
from sklearn.neighbors import KNeighborsClassifier  
from sklearn.svm import SVC  
from sklearn.preprocessing import LabelEncoder, StandardScaler  
from sklearn.model_selection import train_test_split, cross_val_score  
from sklearn.metrics import classification_report, accuracy_score  
import warnings  
warnings.filterwarnings("ignore")  
def load_csv_dataset(path="sleep_health_dataset.csv"):  
    df = pd.read_csv(path)  
    df[["systolic", "diastolic"]] = (  
        df["Blood Pressure"].str.split("/", expand=True).astype(float)  
    )  
  
    gender_map = {"Male": 0, "Female": 1}  
    bmi_map = {"Normal": 0, "Normal Weight": 0, "Overweight": 1, "Obese": 2}  
    disorder_map = {"None": 0, "Insomnia": 1, "Sleep Apnea": 2}  
    df["gender_enc"] = df["Gender"].map(gender_map).fillna(0)  
    df["bmi_enc"] = df["BMI Category"].map(bmi_map).fillna(1)  
    df["disorder_enc"] = df["Sleep Disorder"].map(disorder_map).fillna(0)  
    df = pd.get_dummies(df, columns=["Occupation"], prefix="occ")  
    feature_cols = [  
        "Age", "gender_enc", "Sleep Duration", "Quality of Sleep",  
  
        "Physical Activity Level", "Stress Level", "bmi_enc",
```

```
"systolic", "diastolic", "Heart Rate", "Daily Steps"
]

occ_cols = [c for c in df.columns if c.startswith("occ_")]
feature_cols += occ_cols
X = df[feature_cols].values
y = df["disorder_enc"].values
return X, y, feature_cols

def load_user_reports(path="reports.json"):
    with open(path) as f:
        reports = json.load(f)
    rows = []
    for r in reports:
        try:
            bp_str = str(r.get("bp", "120/80"))
            if "/" in bp_str:
                sys_val, dia_val = bp_str.split("/")
            else:
                sys_val, dia_val = bp_str, "80"
            gender = 0 if str(r.get("gender", "")).lower() == "male" else 1
            bmi_raw = str(r.get("bmi", "")).lower()
            bmi_enc = 0 if "normal" in bmi_raw else (2 if "obese" in bmi_raw else 1)
            risk = str(r.get("riskLevel", "Low")).lower()
            if "high" in risk:
                label = 2
            elif "moderate" in risk:
                label = 1
            else:
                label = 0
            rows.append({
                "Age": float(r.get("age", 30)),
```

```
"gender_enc":      gender,
"Sleep Duration":  float(r.get("sleep_duration", 7)),
"Quality of Sleep": float(r.get("stress", 5)), # map stress as proxy
"Physical Activity Level": 45,                # default

"Stress Level":    float(r.get("stress", 5)),
"bmi_enc":         bmi_enc,
"systolic":        float(sys_val),
"diastolic":       float(dia_val),
"Heart Rate":      float(r.get("heart_rate", 75)),
})
except Exception:
    continue
return rows

if __name__ == "__main__":
    print("Loading and merging datasets...")
    X, y, feature_cols = build_combined_dataset()
    print(f"Total samples: {len(y)} | Features: {len(feature_cols)}")
    print(f"Class distribution — None: {(y==0).sum()}, Insomnia: {(y==1).sum()}, Sleep
Apnea: {(y==2).sum()}")
    clf, scaler, model_name = train_models(X, y)
    print("\n===== Sample Prediction =====")
    sample_user = {
        "age": 34, "gender": "Female", "sleep_duration": 5,
        "stress": 8, "bmi": "Obese", "bp": "127/80",
        "heart_rate": 76, "occupation": "Nurse", "daily_steps": 4000
    }
    result = predict_user(clf, scaler, feature_cols, sample_user)
    print(f"Input : {sample_user}")
    print(f"Predicted Sleep Disorder: {result}")
```

5.2 Output Screenshots

5.2.1 Home Page:

The Home Page of the Sleep Disorder Prediction system acts as the main entry point for users and is designed with a clean, modern, and visually appealing interface. At the top, a navigation bar is provided with options such as Home, About, Register, Login, and Admin, enabling users to easily access different sections of the application. The header clearly displays the title “Classification of Sleep Disorders”, giving users an immediate understanding of the system’s purpose.

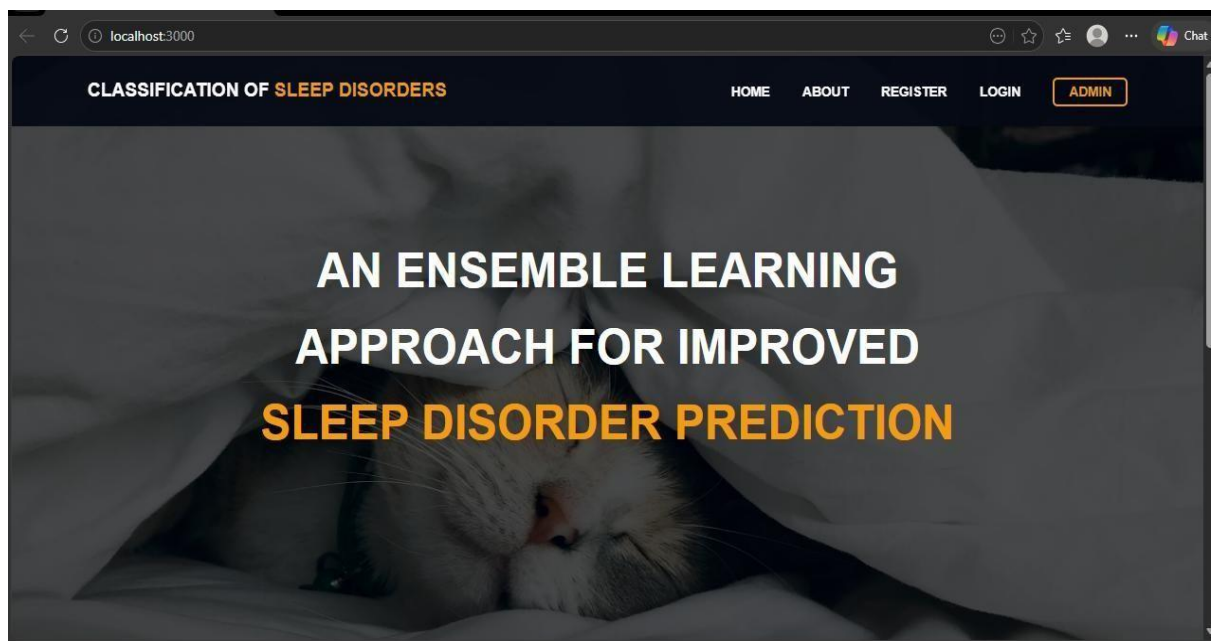


Fig 5.1 Home Page

5.2.2 About Page:

The About Page of the Sleep Disorder Prediction system provides detailed information about the purpose, functionality, and underlying technology of the application. It is designed to help users understand how the system works and the benefits it offers in predicting sleep disorders using advanced machine learning techniques. The page includes a clear heading “About” followed by a descriptive section explaining that the system is a web-based application developed to analyze users’ sleep health based on personal and lifestyle data. Additionally, the About Page highlights key features such as secure user authentication (registration and login), data storage for future reference, and an admin

module for monitoring and managing system records. A relevant healthcare-related image is displayed alongside the text to enhance visual appeal and provide context.

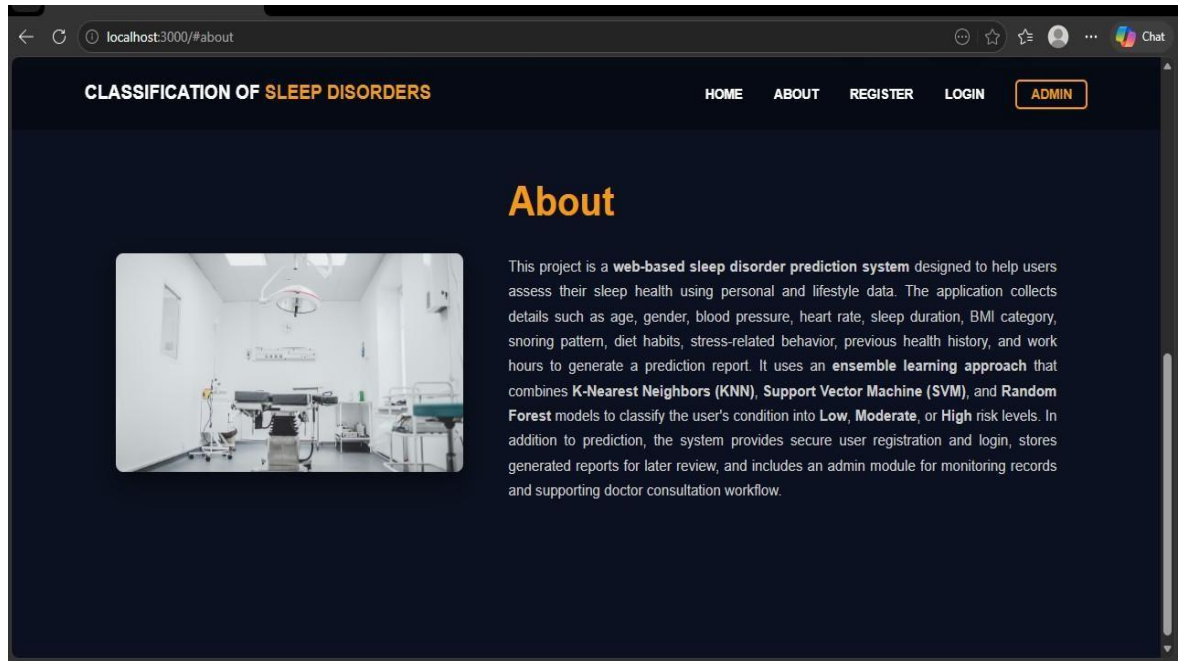


Fig 5.2 About Page

5.2.3 Registration Page

The Register Page of the Sleep Disorder Prediction system enables new users to create an account in order to access the prediction features of the application. It is designed with a simple and clean interface that ensures ease of use and secure data entry. The page contains a centered registration form with clearly labeled input fields, including Full Name, Email Address, Phone Number, and Password.

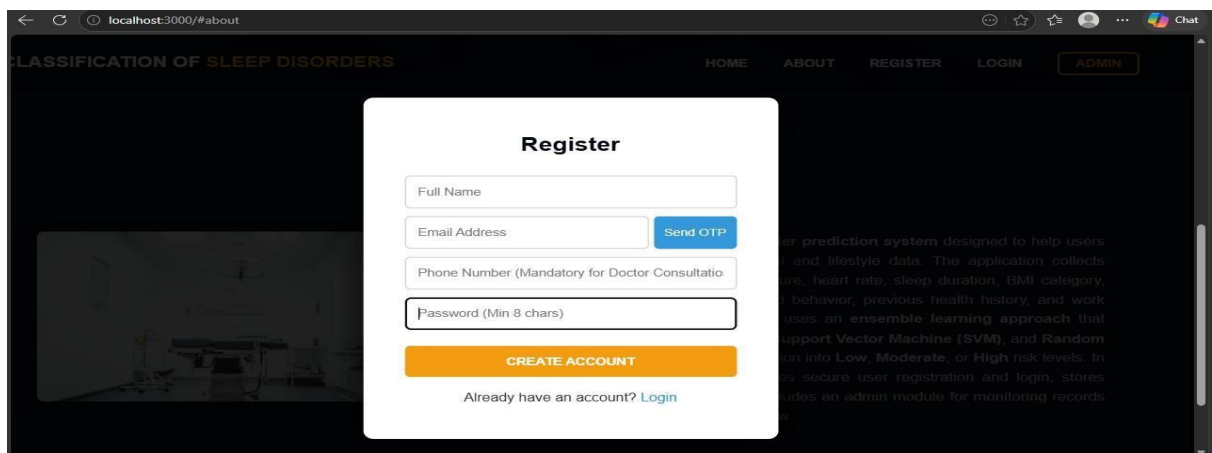


Fig 5.3 Registration Page

5.2.4 : Login Page

The Login Page of the Sleep Disorder Prediction system allows registered users to securely access their accounts and use the prediction features of the application. The page is designed with a simple and user-friendly interface, featuring a centered login form that ensures easy interaction and clarity. It includes essential input fields such as Email Address and Password, which are required for user authentication.

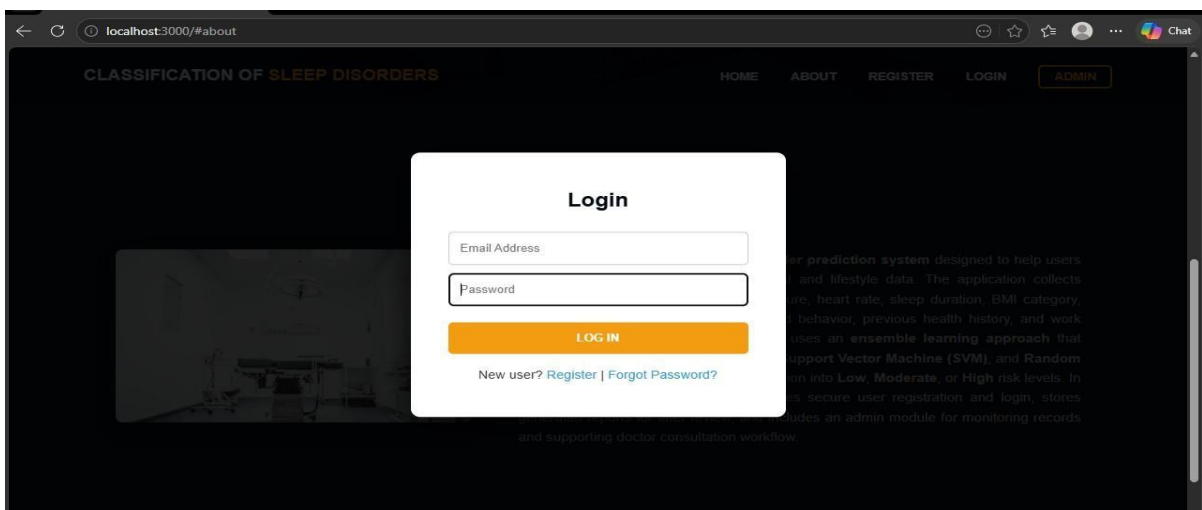


Fig 5.4 Login Page

5.2.5 Prediction Page: This is prediction page. In here, user can input their data and get prediction.

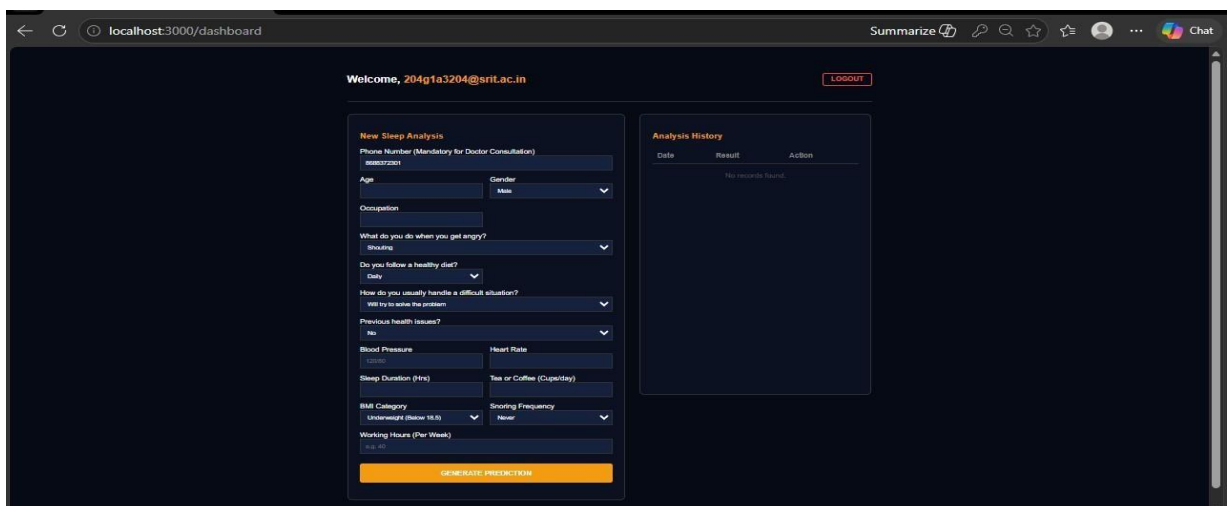


Fig 5.5 Prediction Page

5.2.6. Report Page:

This is the report page. In here result will be display. The Prediction Result Page of the Sleep Disorder Prediction system displays the final output generated after processing the user's input data through the ensemble learning models. This page is designed to present the prediction in a clear, structured, and easy-to-understand format, similar to a medical report.



Fig 5.6 Report Page

5.2.7 Admin Registration Page: This is admin register page. In here Admin can register with their credentials. At the center of the interface, there is a neatly structured registration form enclosed within a glowing green-bordered card, which stands out against the dark background. The heading “**Admin Registration**” is prominently shown at the top, along with a brief subtitle indicating that users can register a new hospital admin account.

The form contains the following input fields:

- **Full Name** – to enter the administrator's name
- **Official Email** – to provide a valid professional email address
- **Password (Min 8 chars)** – ensuring basic password strength
- **Confirm Password** – to verify the password entered

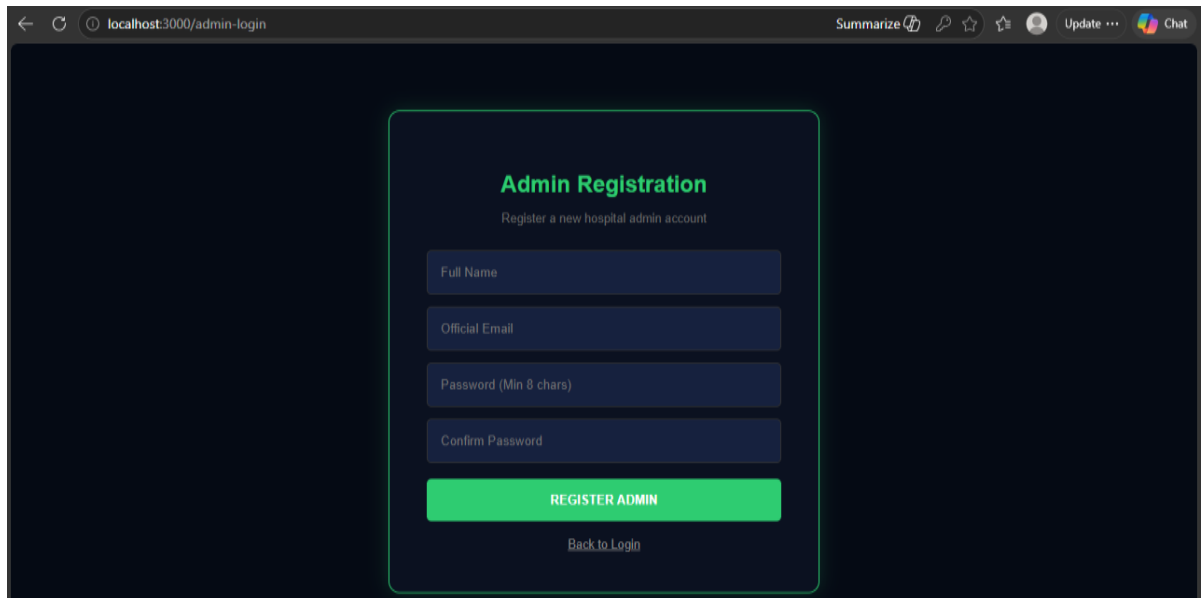


Fig 5.7 Admin Registration Page

5.2.6. Admin Login Page:

At the center of the screen, there is a login form enclosed within a rectangular card highlighted by an orange glowing border, which contrasts clearly against the dark background. The title “Hospital Staff Login” is displayed prominently at the top in bold orange text, indicating that the page is intended for authorized hospital staff members.

The form includes the following input fields:

Official Email (name@hospital.com) – for entering the staff’s registered email address

Password – for secure authentication

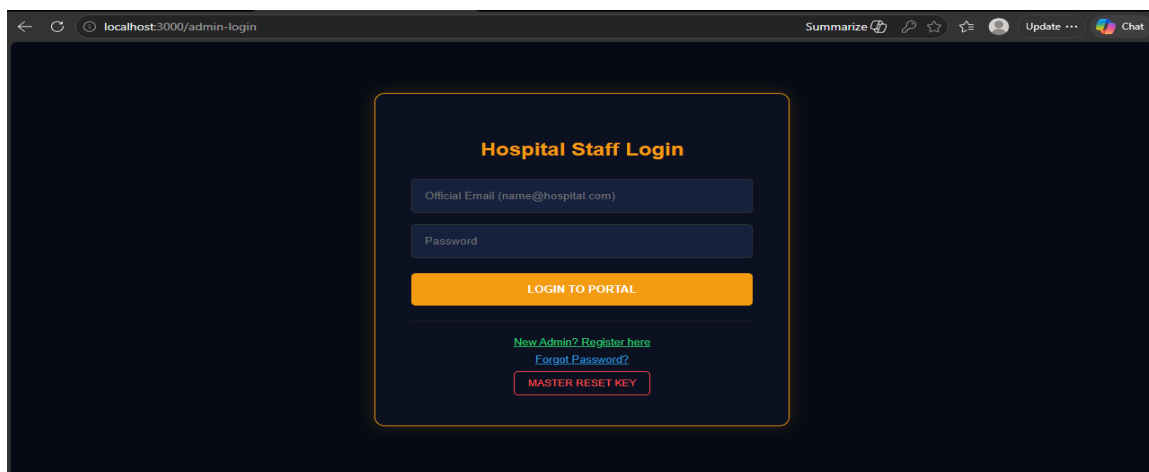
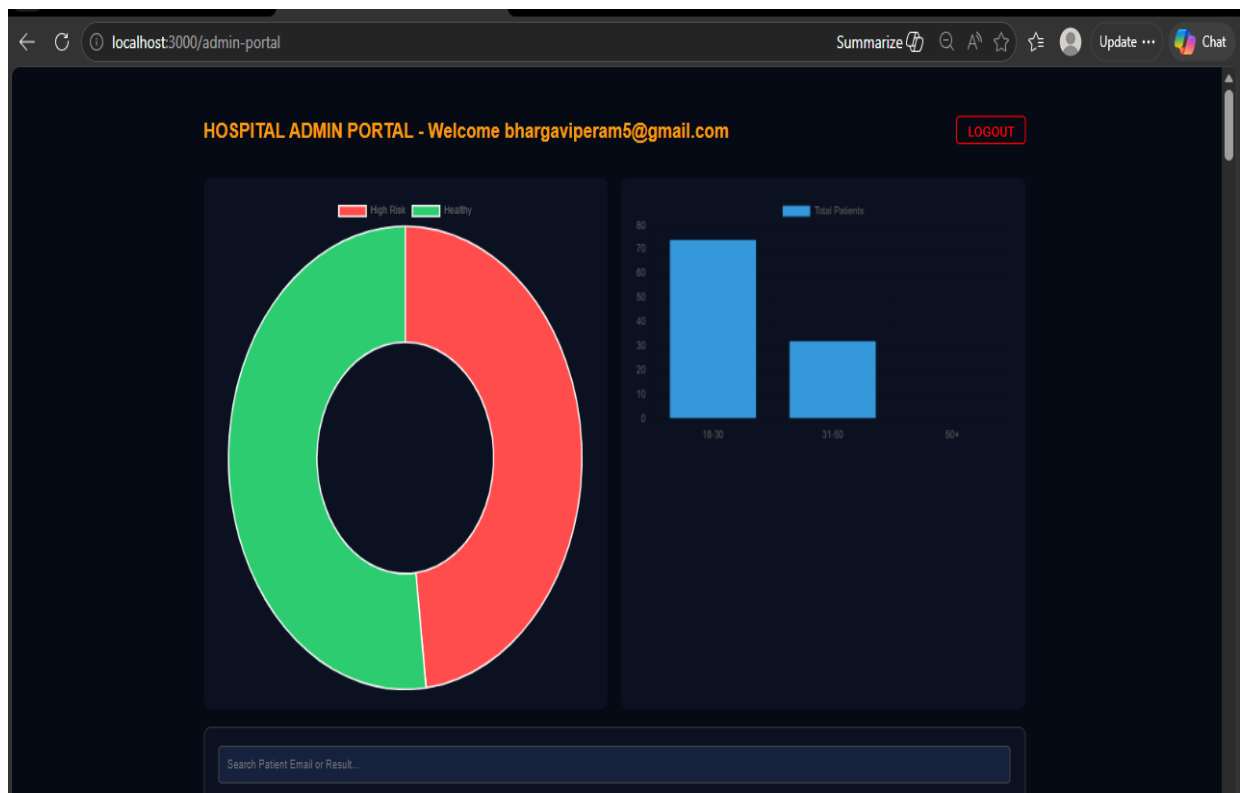


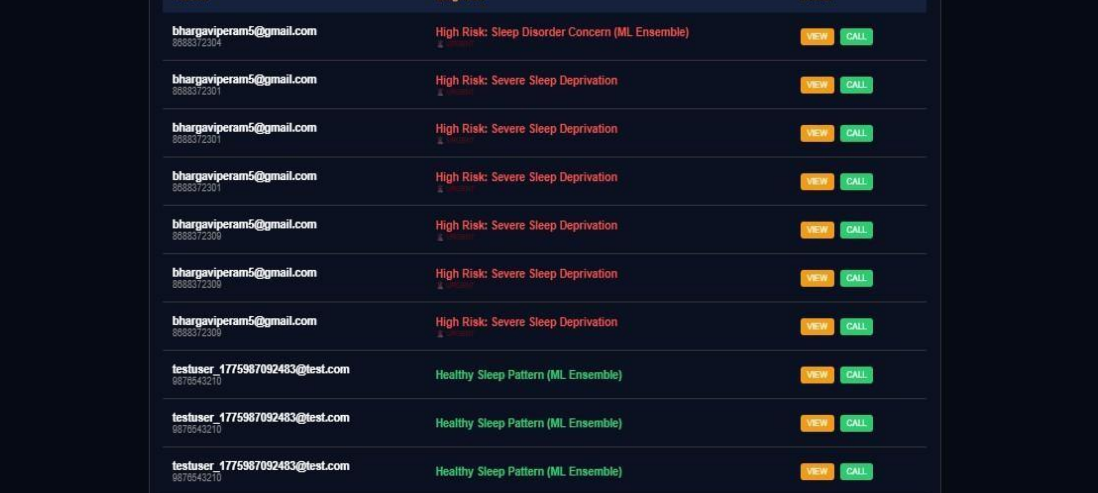
Fig 5.8 Admin Login Page

5.2.7. Admin Page:

This is admin page, admin can view and manage system data and monitor the application. The Hospital Admin Portal Dashboard is a centralized interface designed to assist administrators in monitoring and managing patient health data efficiently. It provides a comprehensive overview of patient statistics and system insights through interactive visualizations and user-friendly components. At the top of the dashboard, a header section displays the portal title along with a personalized welcome message for the logged-in administrator. A logout option is also provided to ensure secure session management.

The dashboard primarily consists of two key analytical components. The first component is a donut chart that visually represents the classification of patients based on their health condition. It categorizes patients into “High Risk” and “Healthy” groups, enabling administrators to quickly identify critical cases that require immediate attention.





The screenshot displays a web application interface for patient management. The browser's address bar indicates the URL is localhost:3000/admin-portal. The interface features a dark theme and includes several navigation elements at the top right: a 'Summarize' button with a magnifying glass icon, an 'Update ...' link, and a 'Chat' button with a speech bubble icon.

Below the navigation bar is a search bar labeled "Search Patient Email or Result...". The main content area contains a table with three columns: "Patient", "Diagnosis", and "Action".

The table lists 12 entries:

- Entries 1-6: Patients with email addresses ending in @gmail.com. Diagnoses include "High Risk: Sleep Disorder Concern (ML Ensemble)" and "High Risk: Severe Sleep Deprivation".
- Entries 7-12: Test users with email addresses ending in @test.com. All have a diagnosis of "Healthy Sleep Pattern (ML Ensemble)".

Each entry in the "Action" column has two buttons: "VIEW" (orange) and "CALL" (green).

Fig 5.9 Admin page

CHAPTER 6

CONCLUSION, LIMITATIONS AND SCOPE FOR FUTURE WORK

6.1 Conclusion

This project develops an efficient system for predicting sleep disorders using an ensemble learning approach that combines multiple machine learning models such as KNN, SVM, and Random Forest. This project develops an efficient system for predicting sleep disorders using an ensemble learning approach that combines multiple machine learning models such as KNN, SVM and Random Forest. By applying Voting classifier the system improves prediction stability and overall reliability compared to individual models. It processes health and lifestyle data through data preprocessing, feature selection, model training, and evaluation to generate meaningful and accurate predictions. The system includes both user and system modules, where users can register, log in, enter patient data, and view results, while the system handles data processing, model execution, and result generation. This approach helps in early identification of sleep disorders, reduces dependency on costly diagnostic methods, and makes the solution more accessible and practical for real-world use. Additionally, the model is scalable and can be extended with larger datasets, more features, and advanced algorithms in the future, making it a strong foundation for improving healthcare decision-making and patient outcomes.

6.2 Limitations

While the Sleep Disorder Prediction System offers an efficient and user-friendly approach for early detection of sleep-related issues, it also has certain limitations. These limitations arise due to factors such as dependency on user-provided data, constraints of machine learning models, and the availability of quality datasets. Understanding these limitations is important to interpret the results correctly and to recognize that the system is intended to support, not replace, professional medical diagnosis.

1. **Dependence on User Input Accuracy:** The system relies heavily on the data provided by users. Incorrect or incomplete inputs can lead to inaccurate predictions.
2. **Limited Medical Validation:** The model provides predictions but cannot replace professional medical diagnosis or clinical tests.
3. **Lack of Real-Time Data Integration:** Without integration with wearable devices, the system may not capture real-time sleep patterns.
4. **Not Suitable for Severe Cases:** The system may not accurately detect advanced or rare sleep disorders.
5. **Privacy and Security Concerns:** Handling personal health data requires strong data protection measures.

6.3 Scope for Future Work

Future enhancements of this project can focus on improving the accuracy and usability of the sleep disorder prediction system by integrating advanced machine learning and deep learning models, including hybrid ensemble techniques. The system can be enhanced by incorporating additional real-time data sources such as wearable devices and sleep tracking applications to provide more comprehensive and accurate inputs. Implementing real-time monitoring and prediction features would make the application more dynamic and useful for continuous health tracking, such as interactive dashboards and personalized recommendations, can enhance user experience. Applying model optimization techniques and expanding the system to detect multiple types of sleep disorders with multilingual support can increase its effectiveness and accessibility to a wider range of users.

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